



Lecture 1: Welcome to CS106S!

Intros, CS for Social Good, JavaScript

Ben Yan, September 24th, 2026

Slides at **cs106s.stanford.edu** :)

[PDF of this presentation](#)



Stanford | ENGINEERING
Computer Science



Welcome to our first day of class!

Hope that you've had a beautiful first week so far!



So glad to be teaching
this 1-unit wonder again!

Like GTA, this is the 6th edition!



Ben Yan
Instructor



Prof. Jerry Cain
Faculty Sponsor



May have been put on hiatus for a year oops
(probably due to catastrophic spending on boba 🙄)

**So glad to be teaching
this 1-unit wonder again!**

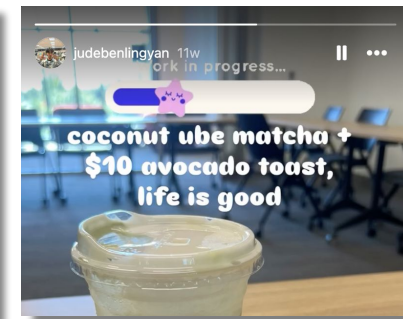
Like GTA, this is the 6th edition!

Did I say I was
retiring two years
ago? Maybe oops :)

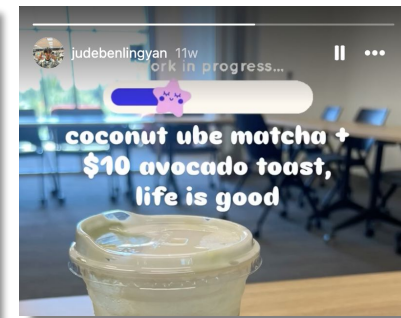
Seriously, who lets
me still do this?

It was deadass a reel on
Instagram that inspired
me to teach again! Like I
really miss this j*b 😭





So thank you for being here!!
And I hope to make your time fun and well worth it ❤️



So thank you for being here!!
And I hope to make your time fun and well worth it ❤️

But in case it's not, the add-drop deadline is Week 3, Friday, 5 PM!



No hard feelings if you drop! I, in fact, aaahh dropped this class as a frosh many years ago—yikes. Not exactly a ringing endorsement...



Alrighty, let's do introductions...





Hi I'm Ben! (aka Disastrous Life of Benji Y.)



- ❑ I'm a **CS MS/coterm**! (an euphemism for not knowing tf I'm doing with my life lol)
- ❑ Undergrad was an absolute trainwreck: nearly triple majored 🙄 in **CS, Math, English (Creative Writing)**, and after much drama, ended up with a double major and minor.
- ❑ A few **side quests**: studied at Oxford, was in a Shakespeare play, wrote 2 bad novels **Writing and teaching** is what I love the most. have section-led / TAed for **CS106A, 106AX, 106B, 107**, and **Math 19/20/21/51**. This is my 11th time teaching for Jerry!
- ❑ Head TA for CS106AX this autumn, but also **teaching this for the love of the game!**

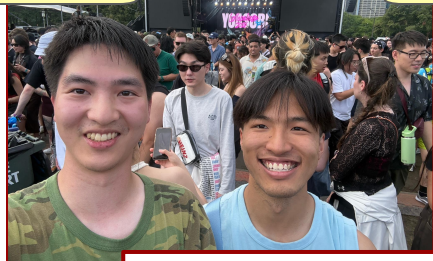




Hi I'm Ben! (aka Disastrous Life of Benji Y.)



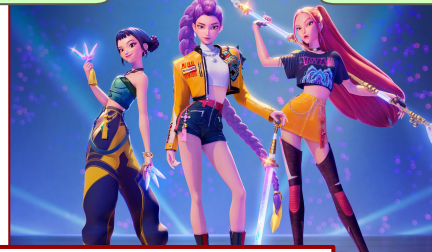
YOASOBI J-pop
(Novels into Music!)



Manga and Anime



K-pop and BTS!



What also brings me passion and joy in life!!

한글 日本語
漢語 漢語



Language
Learning



Food, especially when
I don't pay for it :)



And of course
Stanford yeah!??

But more importantly, introduce yourselves! 🤝

First, share the following **just with your table group** / people around you!

But more importantly, introduce yourselves! 🤝

First, share the following just with your table group / people around you!

- ❑ Your name!
- ❑ Your school year
- ❑ What you're studying / thinking about studying
- ❑ **Icebreakers: share a fun fact about yourself!**

Some ideas for fun facts :) ↙



An unusual or special hobby that you enjoy



Something fun you did over the summer break?



Music, albums, books, shows, movies you love?

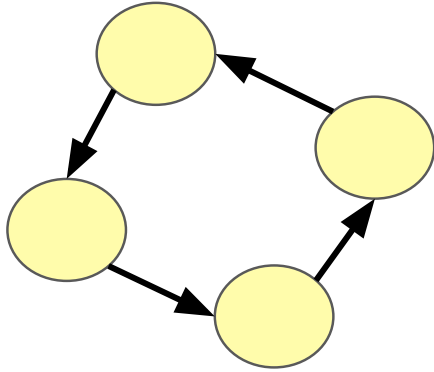


What classes are you taking? Any drama?

Or anything else you'd like to share!



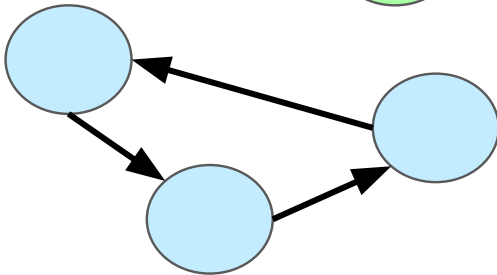
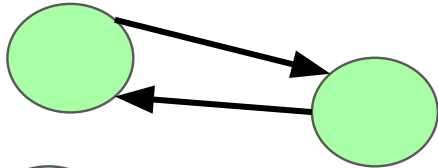
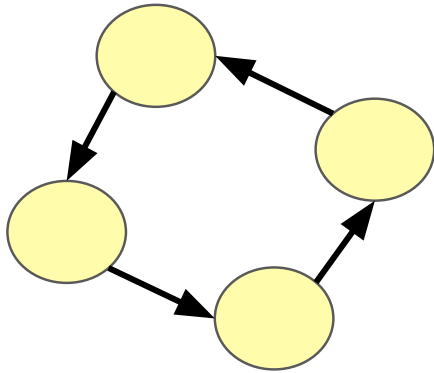
Next, we'll do a "Josephus circle" of intros!



At each table, each person introduces—to the whole class—the person to their right! Share 1-2 sentences / a biography of their liking



Next, we'll do a "Josephus circle" of intros!



At each table, each person introduces—to the whole class—the person to their right! Share 1-2 sentences / a biography of their liking



You can also pair up or do a group of three, and introduce each other! I don't mind how you choreograph if it works and everyone's in the fun.

You may want some extra time to remember / jot down each other's facts, and that time begins now! :)



Awesome, thank you all for sharing, phew!



I'll make it my mission to learn all your names (and your interests / quirks you'd like to share!) by next week, so thank you for your patience in the meantime!!



Today's Plan (Usual first day stuff)

1

Syllabus, Logistics,
ask any questions!

3

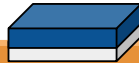
HTML, CSS, group work
on a JavaScript Tutorial!

2

Getting Setup to
Code in JavaScript!

4

Group coding with some
cryptography exercises



Our Classroom Values (From Chris Piech's CS109)

Everyone is welcome.

Intellectual Joy.

Be kind.

Be humane.



Social connection.

Learn by doing.

Thrill of building.

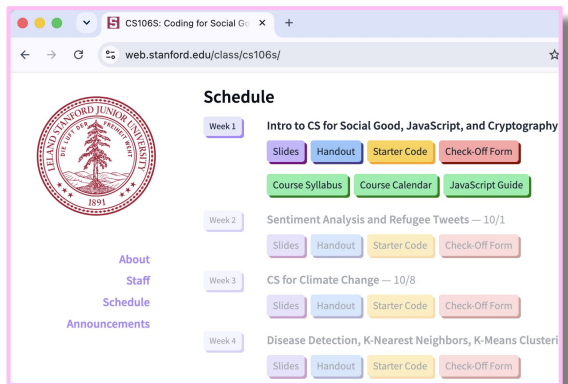
Adapt to new contexts.



Any questions before we
jump into syllabus stuff?



cs106s.stanford.edu



All materials (slides, handouts, code, solutions) will be posted to the website, not Canvas!

cs106s.stanford.edu

This website is your best friend here. You'll want to pay it a visit at the start of each class!



Course Mechanics!



**CS106S: Coding
for Social Good**

- ❖ **1 unit wonder, S/NC**
- ❖ **Attendance (7/9*), no
work outside of class!**
- ❖ **Never failed a student :)**

***Please do reach out if difficult
circumstances arise! I understand life
can be stressful and challenging, and will
always create a path for you to pass ❤️.**

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Course Website!

`cs106s.stanford.edu`



Contact Email

`bbyan@stanford.edu`, or for some reason,
`judebenlingyan@stanford.edu`



Place & Time

Lathrop Building, Room 190

Thursday, 4:30 - 6:20 PM

But I usually keep class to 90 minutes!



Office Hours

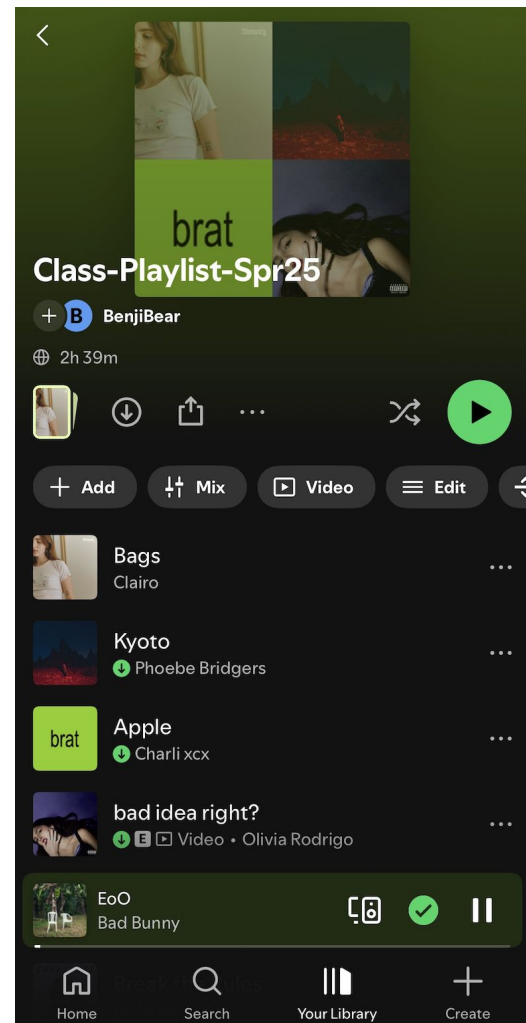
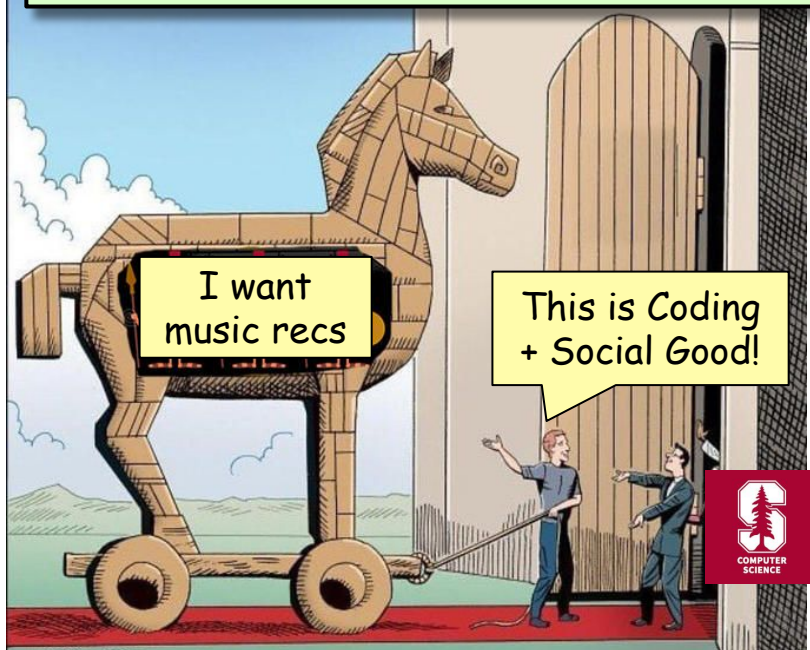
After class, or email/Slack me!

No assignments in 106S, but happy to chat
about material, CS, Stanford, anything!



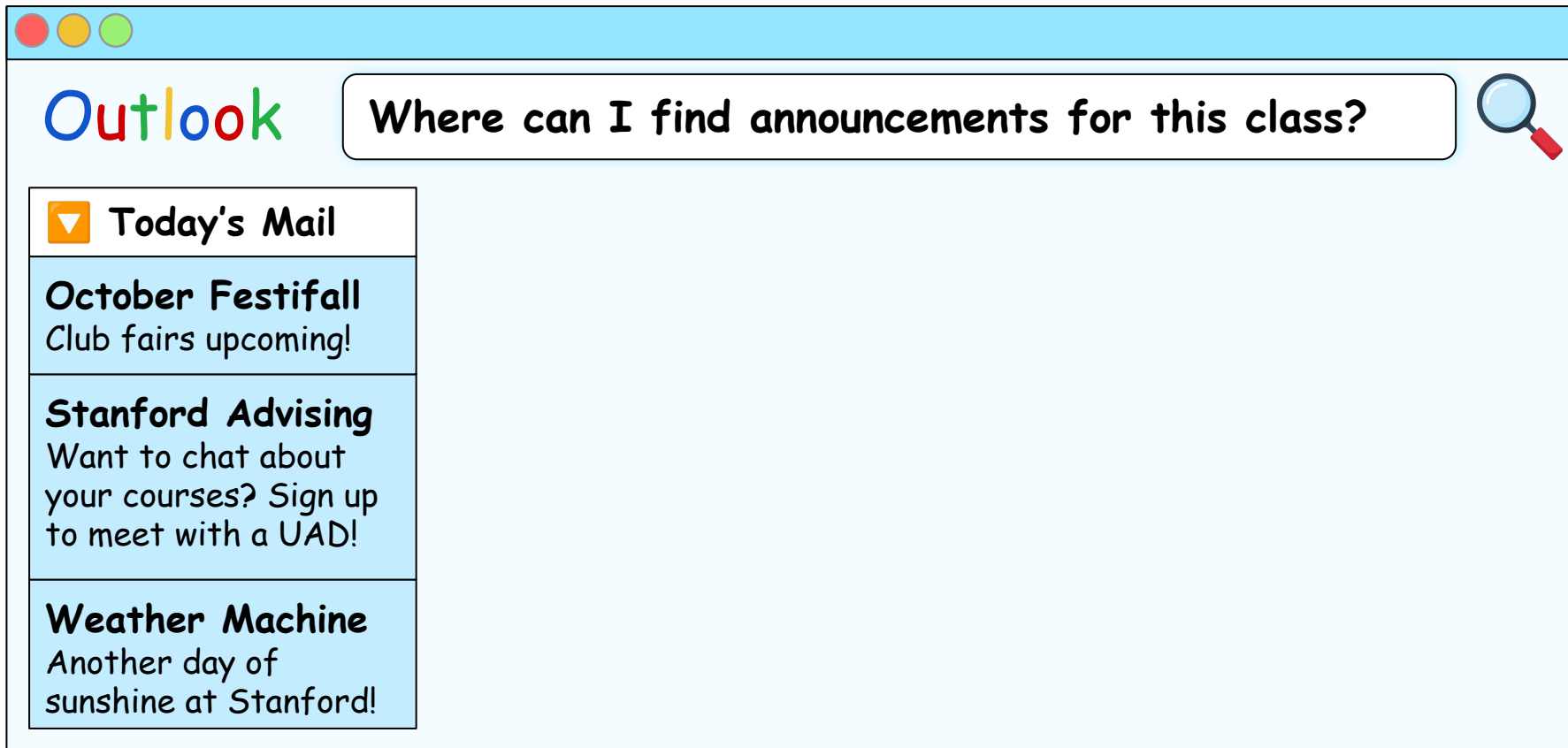
Course Music Playlist!

Most of our class time is relaxed group work! I usually play background music, and **feel free to tell me / submit your song recs** on the check-off form!





Course Mailing List - Make sure you're signed up!





Course Mailing List - Make sure you're signed up!

Where can I find announcements for this class?

Today's Mail

October Festifall
Club fairs upcoming!

Stanford Advising
Want to chat about
your courses? Sign up
to meet with a UAD!

Weather Machine
Another day of
sunshine at Stanford!

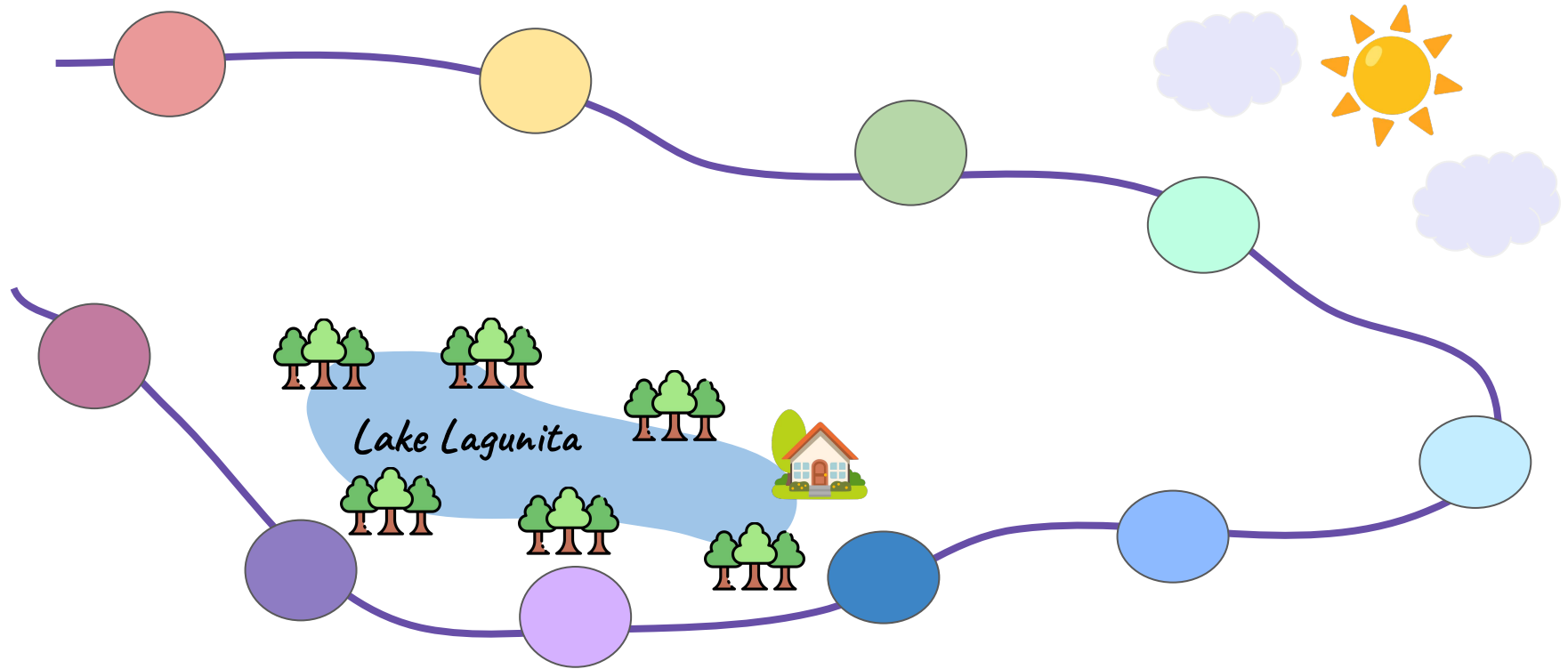
**cs106s-aut2627-students
@lists.stanford.edu**

I'll **broadcast announcements and gossip** about once a week **to your emails via this mailing list**. If you're on it, you've already received a **Welcome Email** from me! If you didn't receive it, let me know, and I'll add you! :)

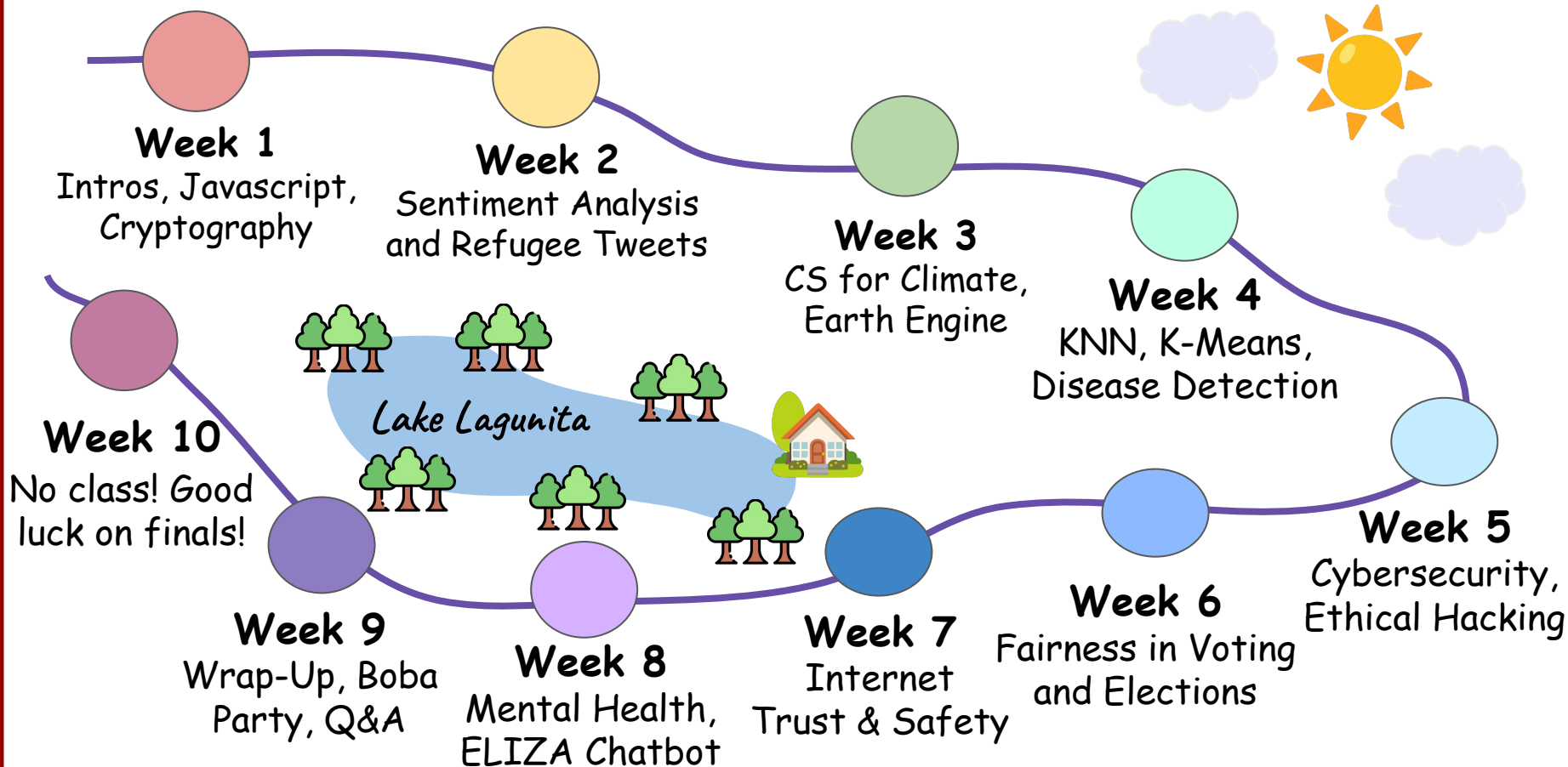
Next, let's talk about
what's on the calendar!



Campus tour of topics this autumn - subject to change!



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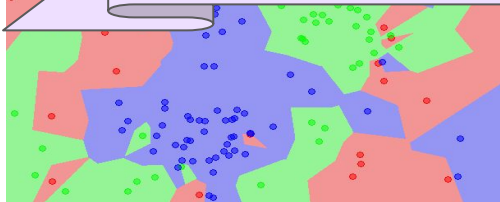
Overview of Class Content!

Coding for Social Good



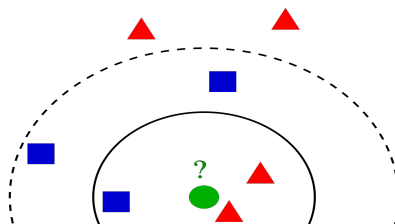
Overview of Class Content!

What **technologies** (machine learning, safety checking, etc.) can be used for **positive impact**?



Coding for Social Good

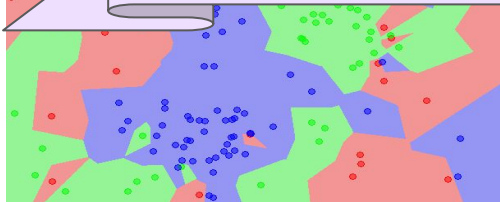
How can we use **algorithms** to materialize ideas into **real-world applications**?



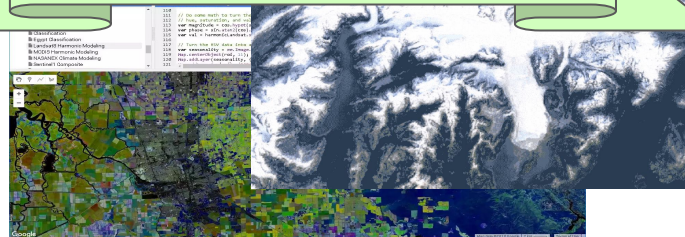


Overview of Class Content!

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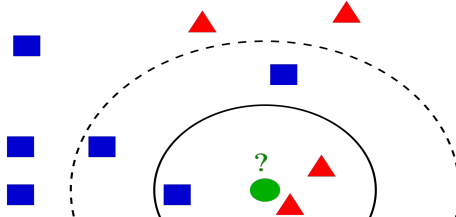


In what areas (e.g., climate, health, user safety) can we use coding for positive impact?

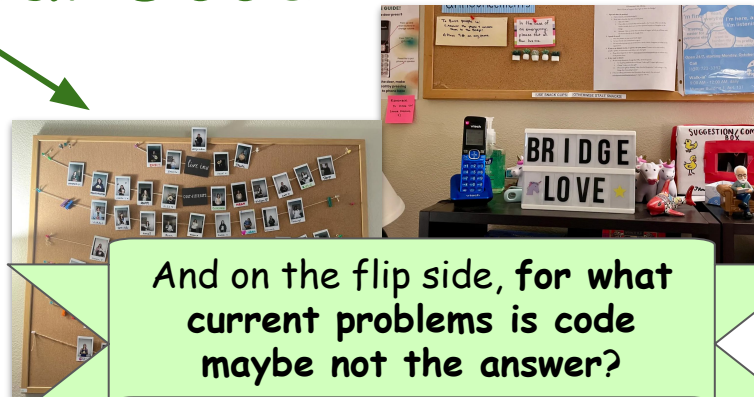


Coding for Social Good

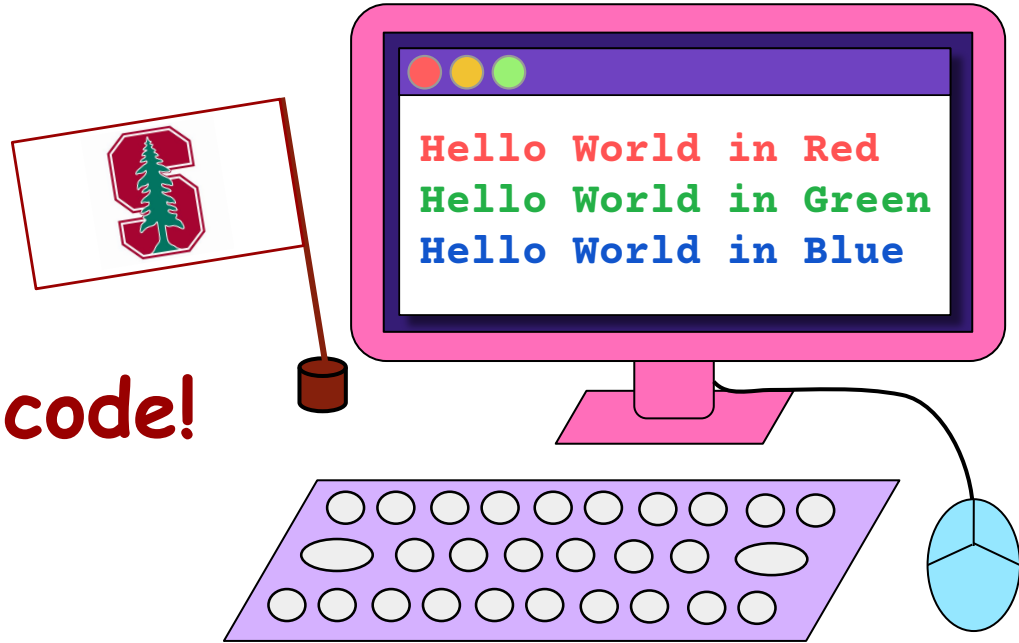
How can we use **algorithms** to materialize ideas into **real-world applications**?



And on the flip side, for what **current problems** is code maybe not the answer?



Let's get setup to code!

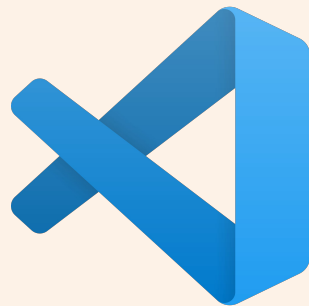




install Chrome

(Safari should also work just fine!)

Getting Set Up

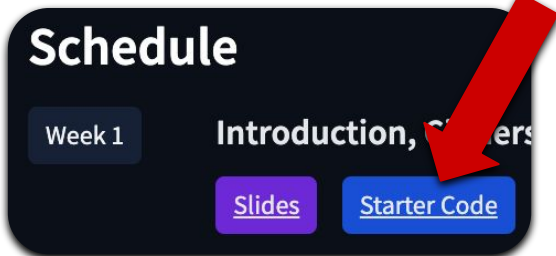


install VS Code

(or an editor of your choice, Sublime Text is also great!)



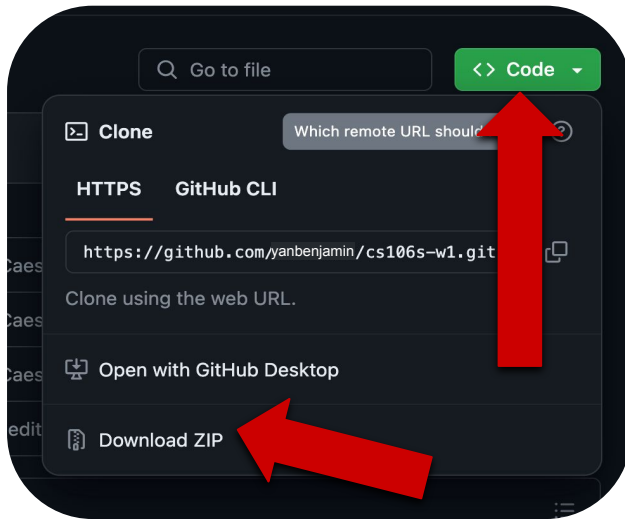
Opening the Starter Code!



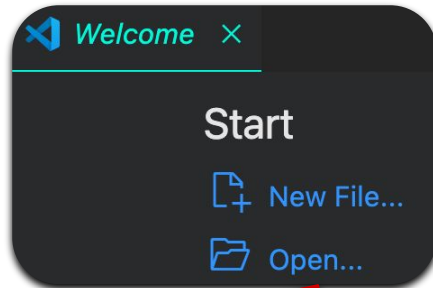
- 1 Navigate to Week 1 of the Schedule section of cs106s.stanford.edu

Also, at this link:

<https://github.com/yanbenjamin/cs106s-w1>



- 2 Click the bright “Code” button, then click “Download ZIP”



In VSCode

- 3 Unzip the download (clicking .zip file should do the trick) and open the folder / files in your editor



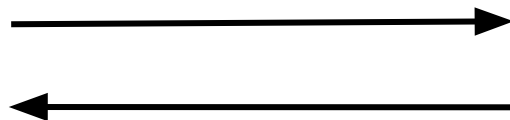
So ... what is `index.html`?

- ❑ In the starter code, you'll find a file named `index.html`; using File Finder or your OS equivalent, open it in Google Chrome!
- ❑ This is the **homepage** of a website, e.g., `www.stanford.edu`.



Client Browser

HTTPS Request for
`cs106s.stanford.edu`



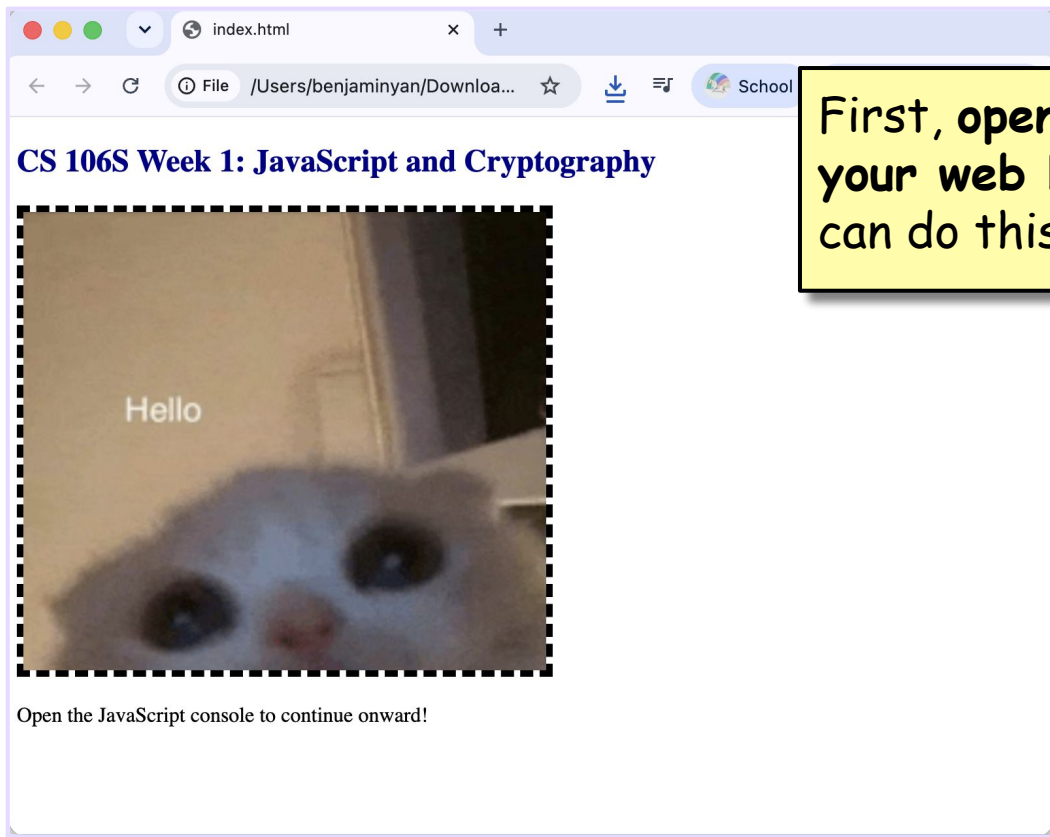
HTTPS Response of
`index.html`



Web Server



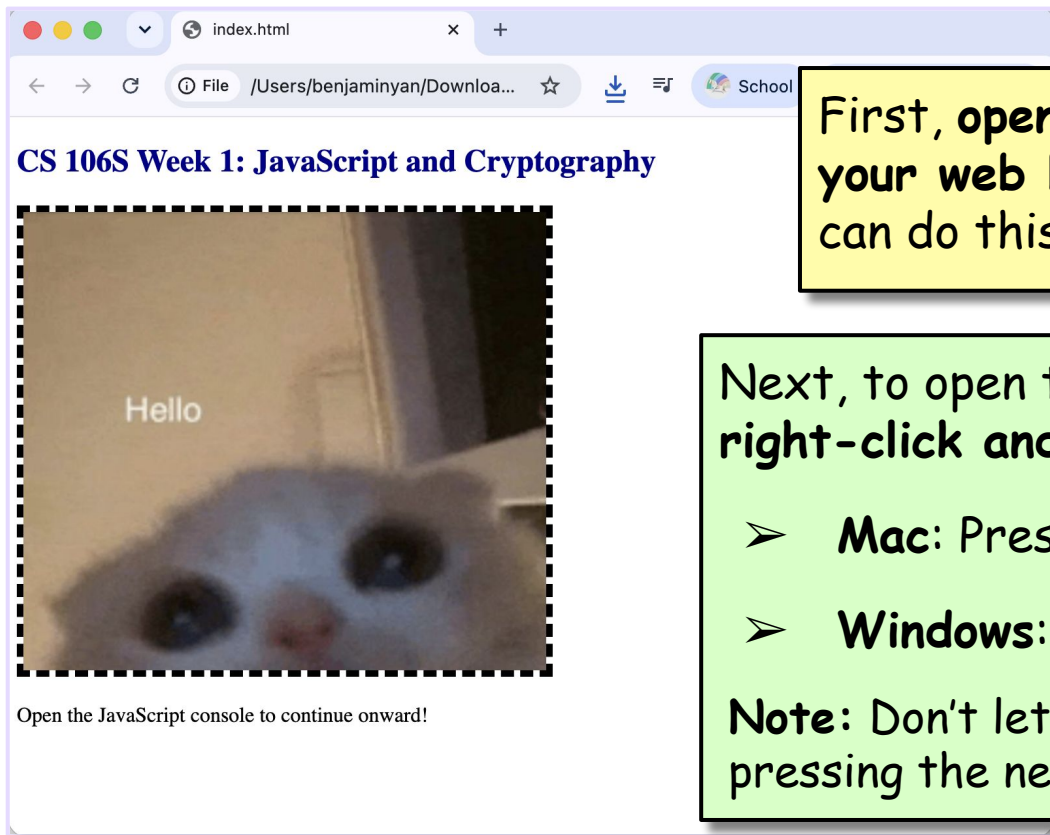
JavaScript in the Browser (Chrome)



First, open the `index.html` file in your web browser, e.g., Chrome! You can do this via Finder / File Explorer.



JavaScript in the Browser (Chrome)



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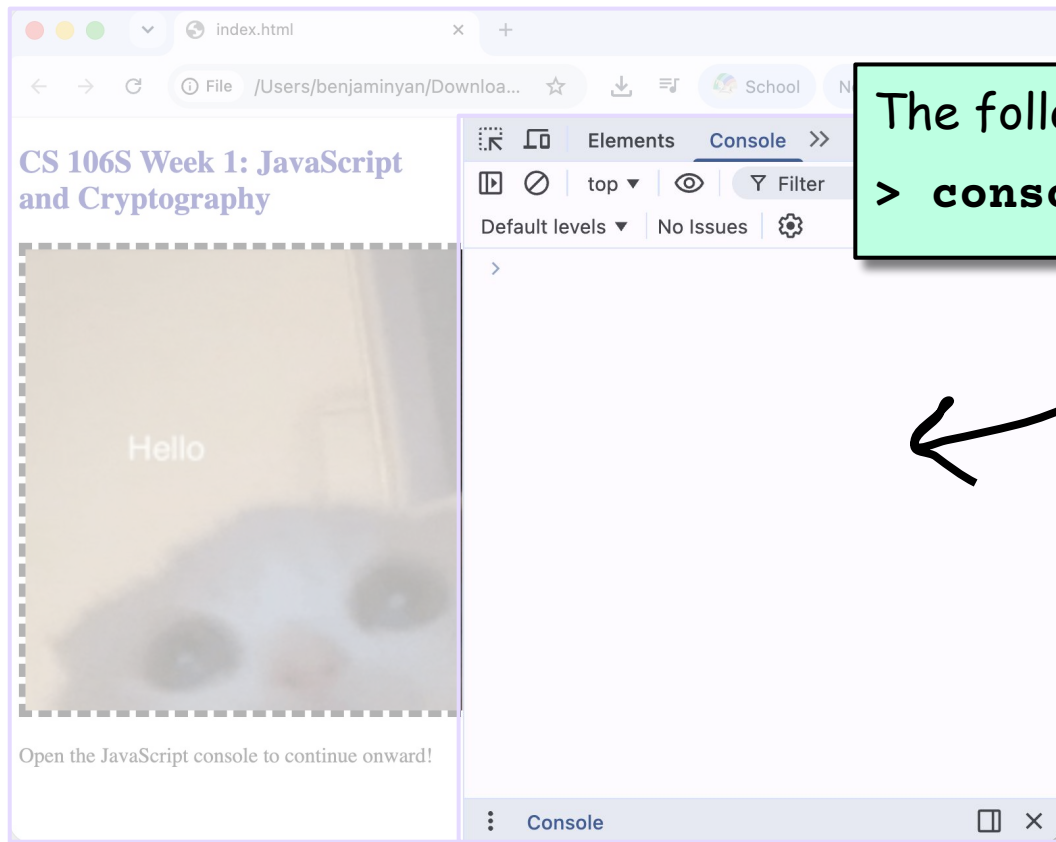
Next, to open the JavaScript console, either right-click and click **Inspect (Element)**, or

- Mac: Press `cmd` — `option` — `j`
- Windows: Press `ctrl` — `shift` — `j`

Note: Don't let go of the previous key while pressing the next key!

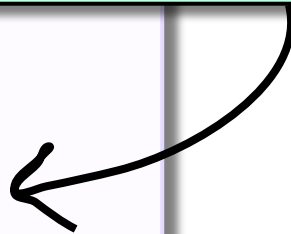


JavaScript in the Browser (Chrome)



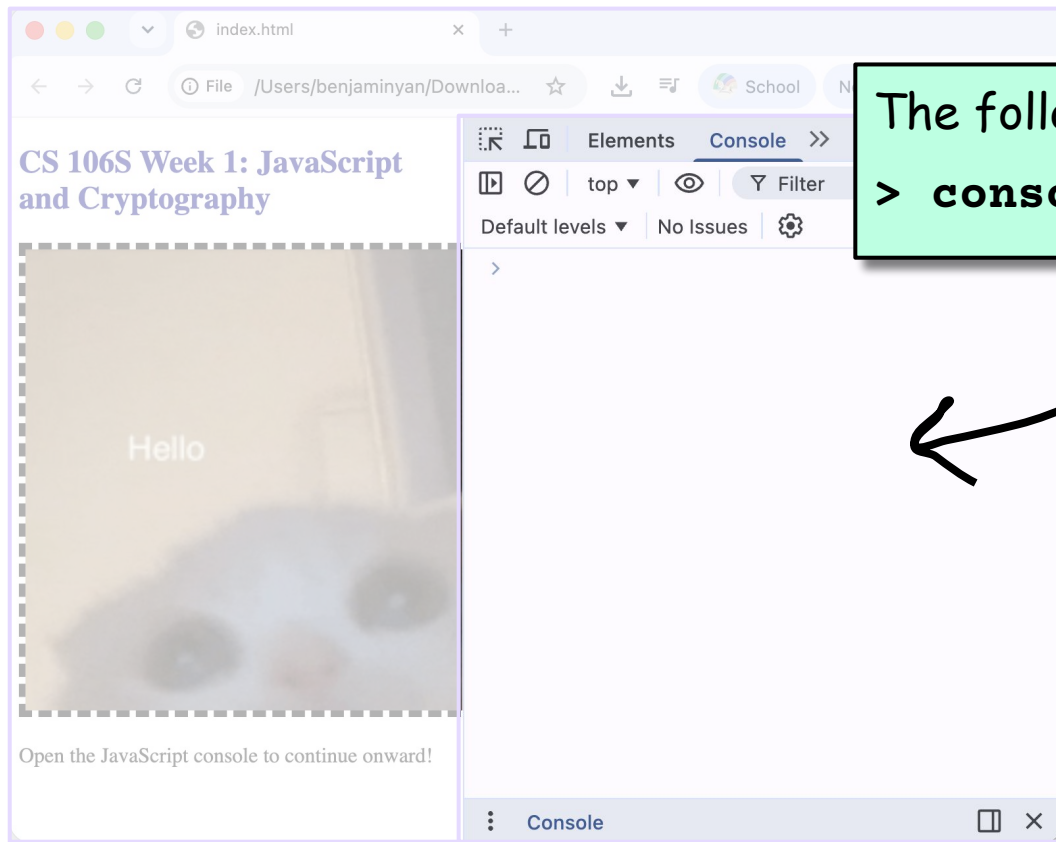
The following panel should pop up!

```
> console.log("Hello World!")
```





JavaScript in the Browser (Chrome)



The following panel should pop up!

```
> console.log("Hello World!")
```

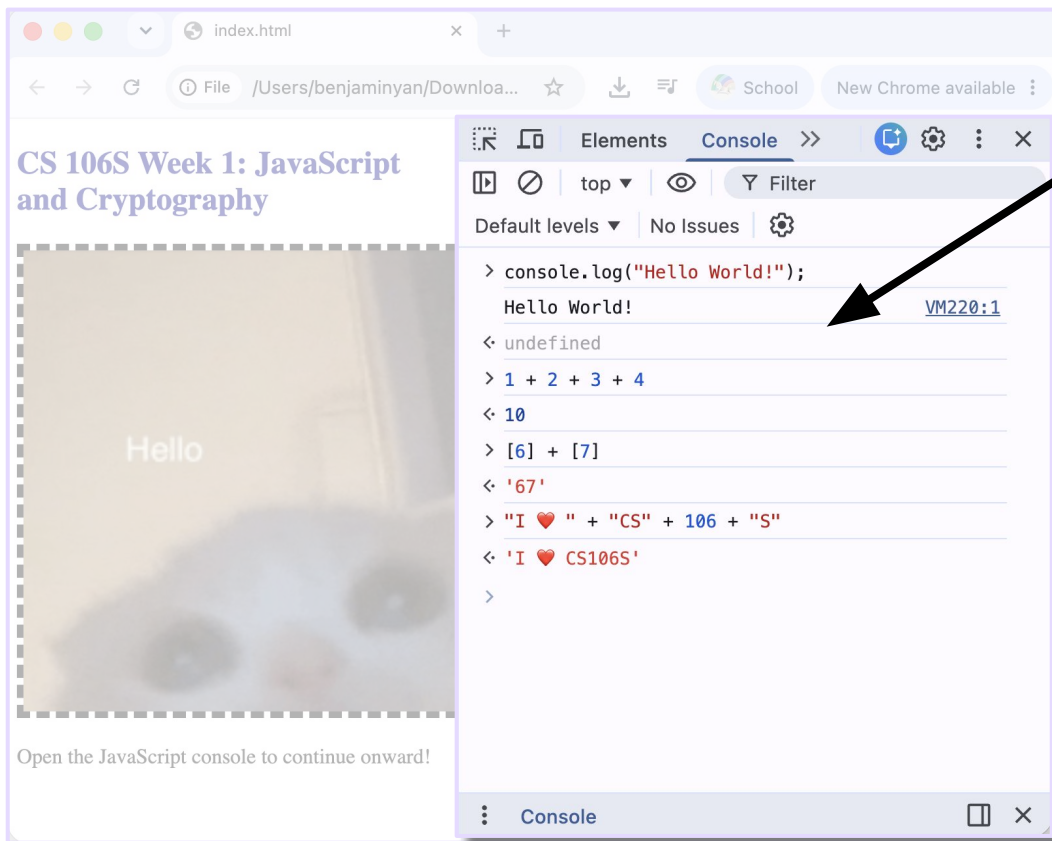
Like a Python interpreter, you can type basic commands, and press **ENTER** to have them run.

Let's go try it out!! :)

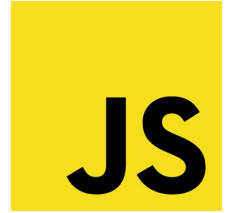


JavaScript in the Browser (Chrome)

Here are some fun snippets of code I ran!



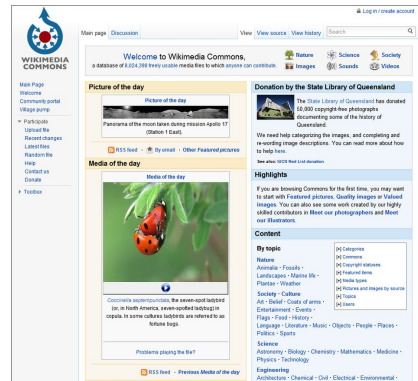
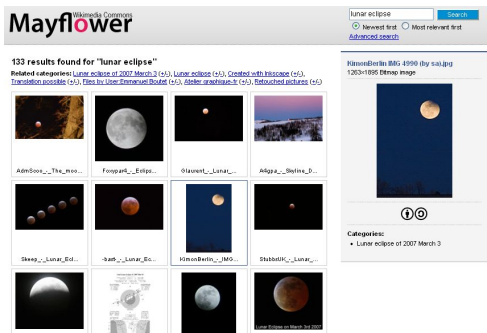
So, what's in this
pandora box of files I
just made you open? 🐱





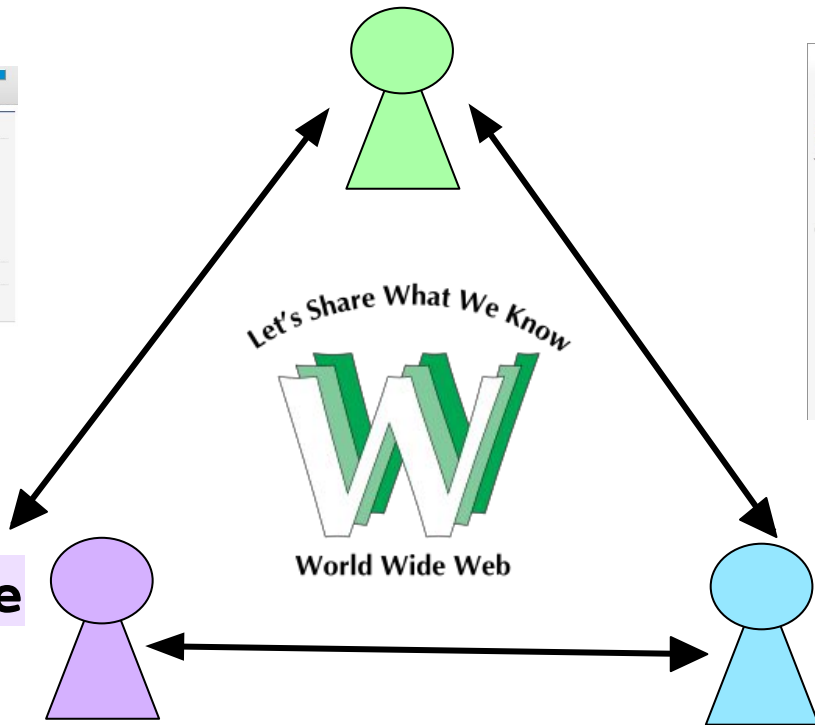
Web Development Love Triangle Yaoi/Yuri

Hypertext Markup Language (HTML)

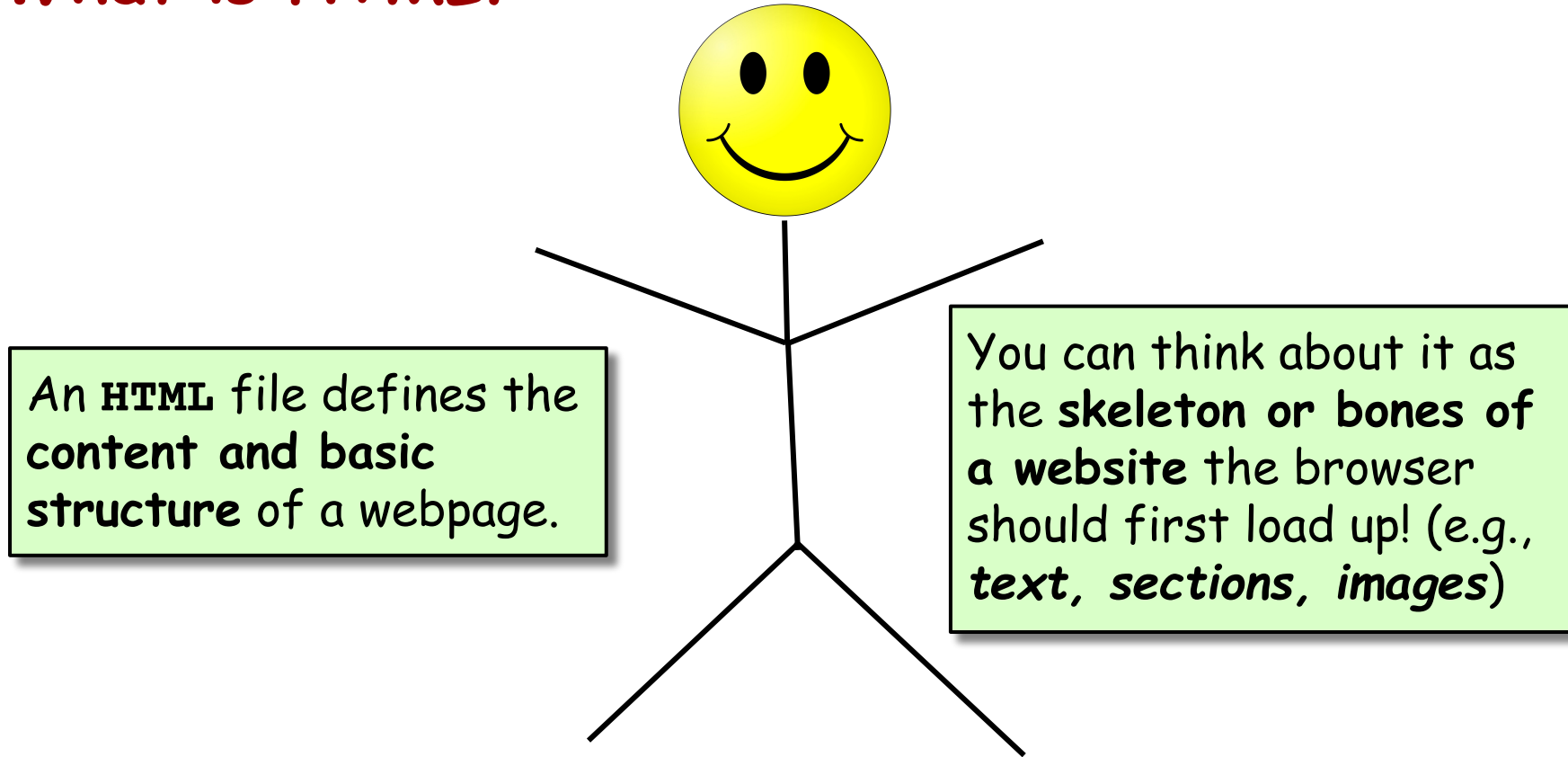


**Cascading Cycle
Sheets (CSS)**

**JavaScript
(JS)**

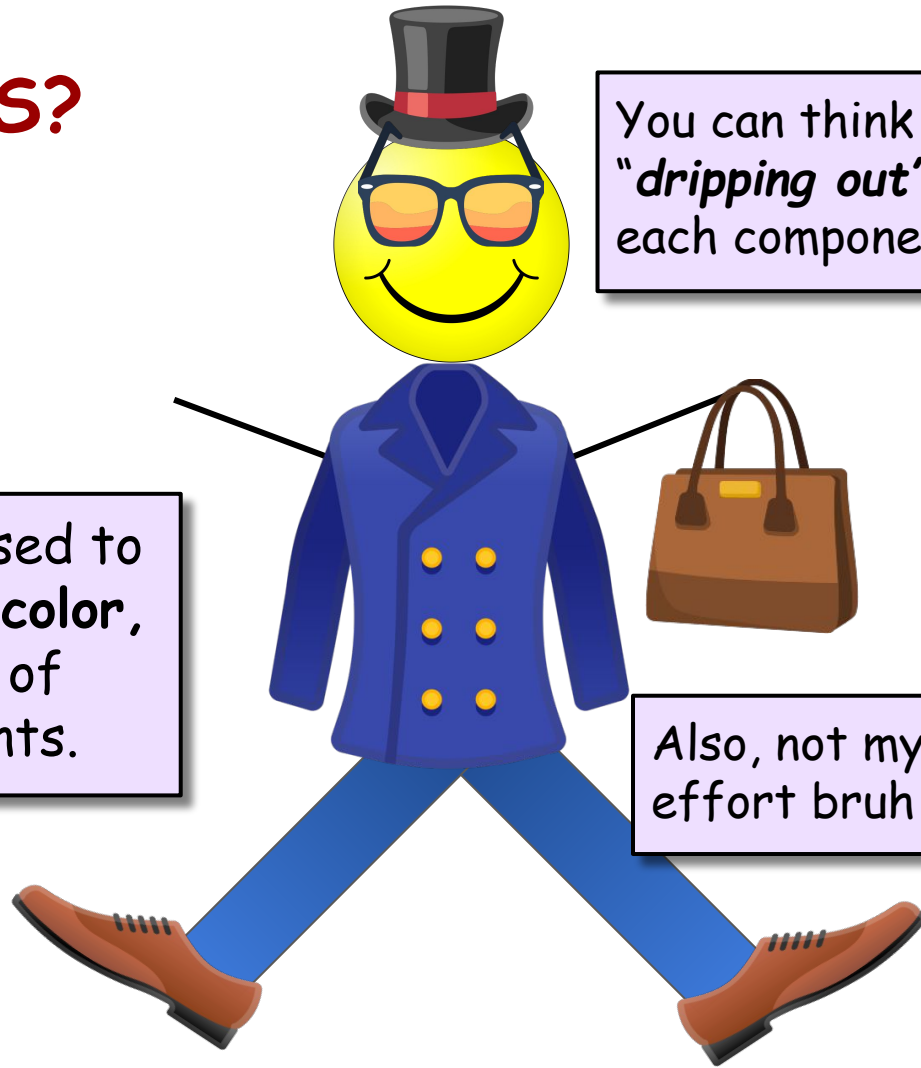


What is HTML?



A stick figure metaphor

What is CSS?



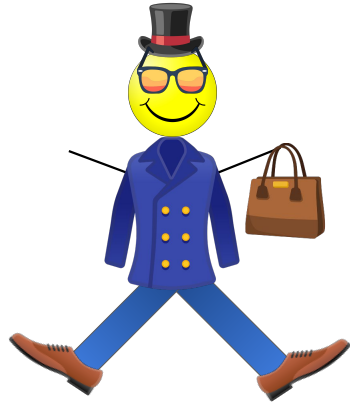
You can think about it as "*dripping out*" the webpage, so each component looks good :)

An `css` file is used to regulate **style**, **color**, and **formatting** of HTML components.

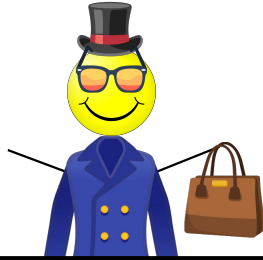
Also, not my best Powerpoint effort bruh **wtf** is this.

¿Cual es JavaScript?

JS (JavaScript) files are the engine that enable the **HTML/CSS** components to be truly interactive, i.e. like a true modern webpage :)



¿Cual es JavaScript?



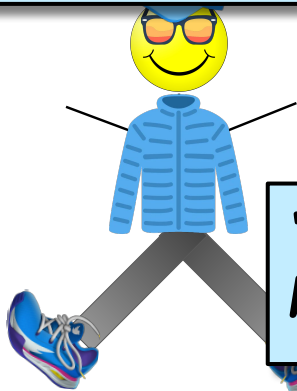
JavaScript is why webpages can respond programmatically to the user, e.g., mouse and key clicks.



"Celebrated" as "the language of the web"

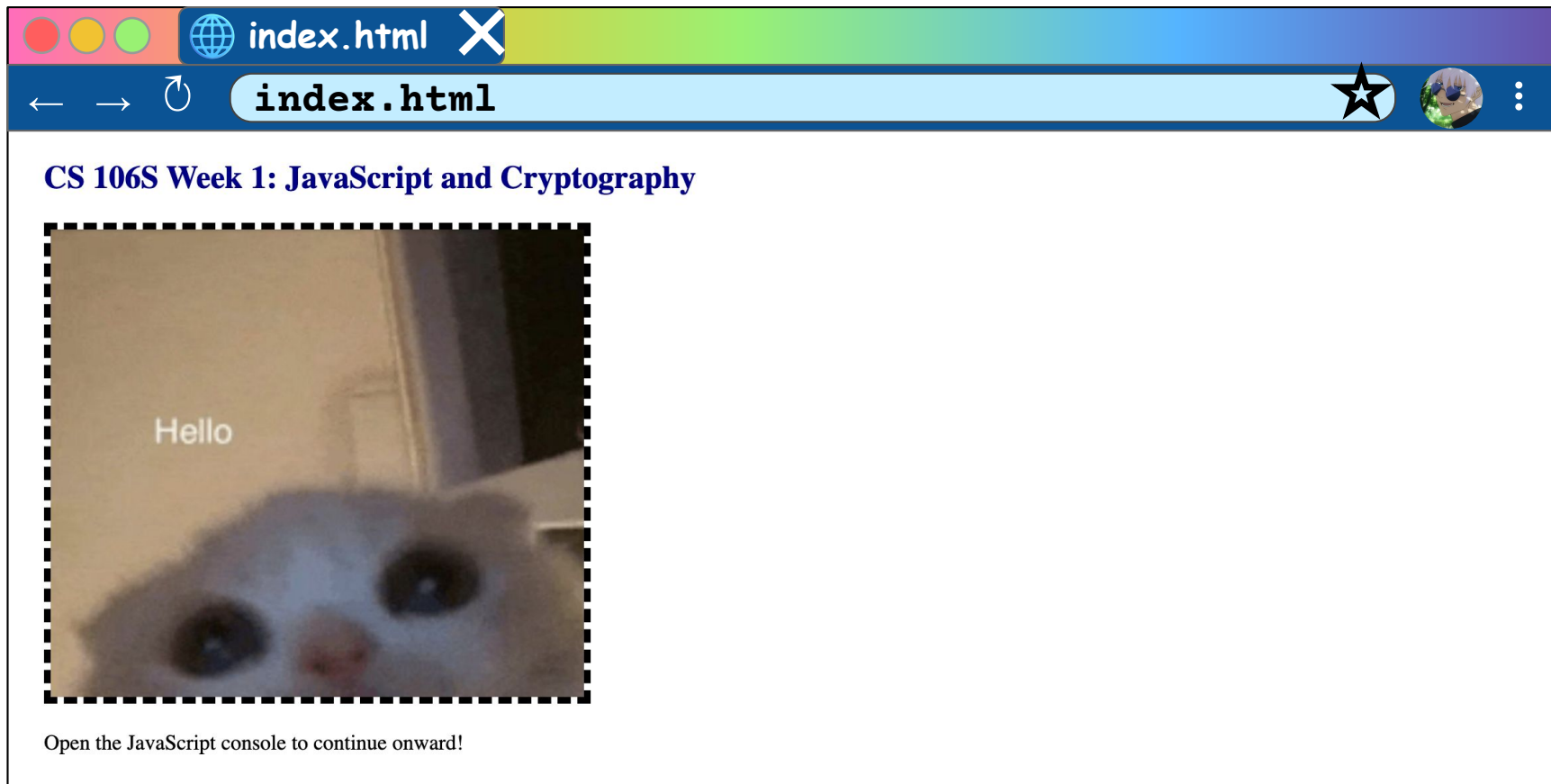
90% of websites use Javascript on the client side, making it essential for browser applications!

JS (JavaScript) files are the engine that enable the **HTML/CSS** components to be truly interactive, i.e. like a true modern webpage :)



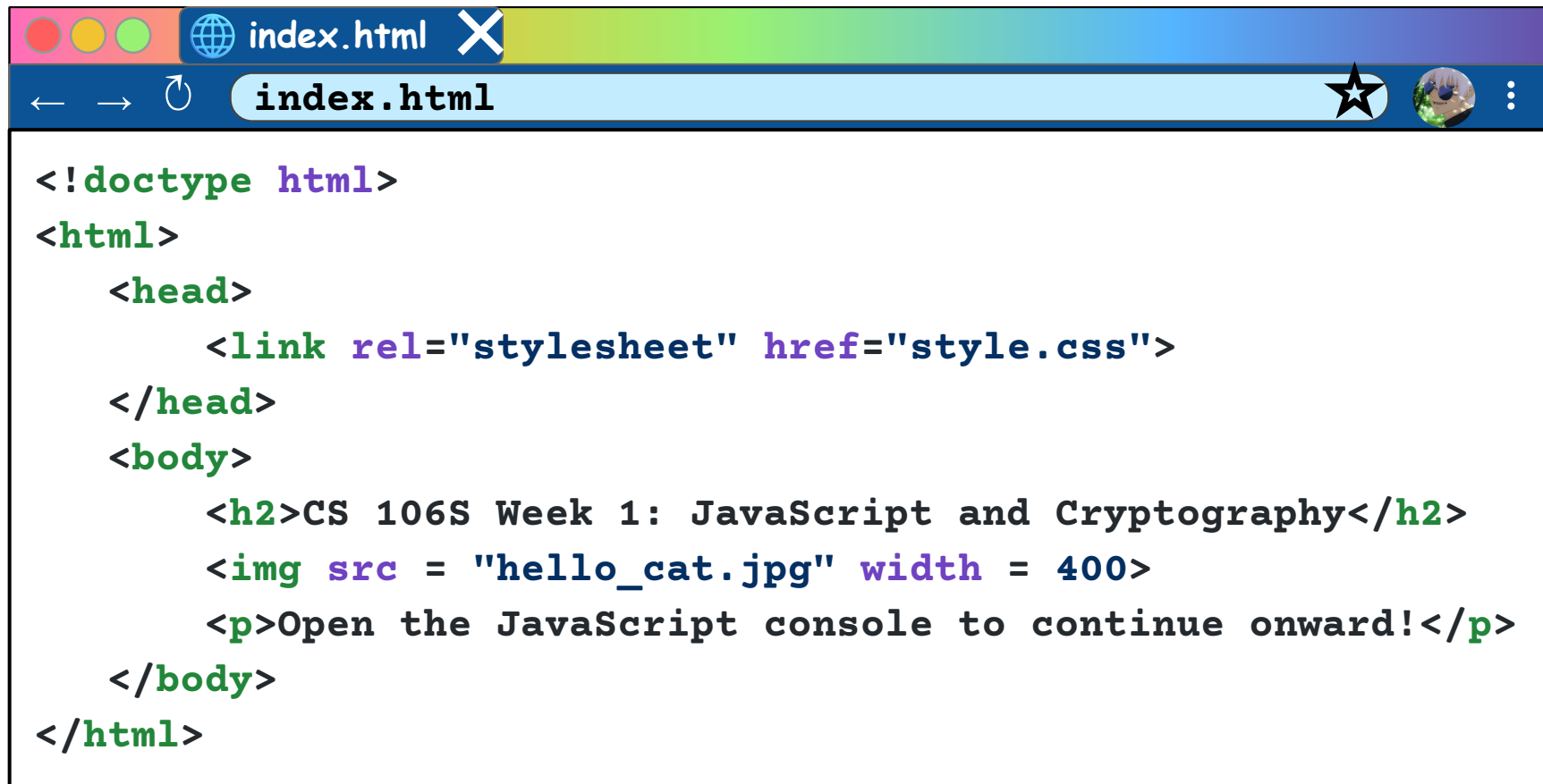


Anatomy of an HTML File





Anatomy of an HTML File



```
<!doctype html>
<html>
  <head>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h2>CS 106S Week 1: JavaScript and Cryptography</h2>
    <img src = "hello_cat.jpg" width = 400>
    <p>Open the JavaScript console to continue onward!</p>
  </body>
</html>
```



Anatomy of an HTML File

 The `<html>` and `</html>` tags enclose all the content.

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Anatomy of an HTML File



index.html

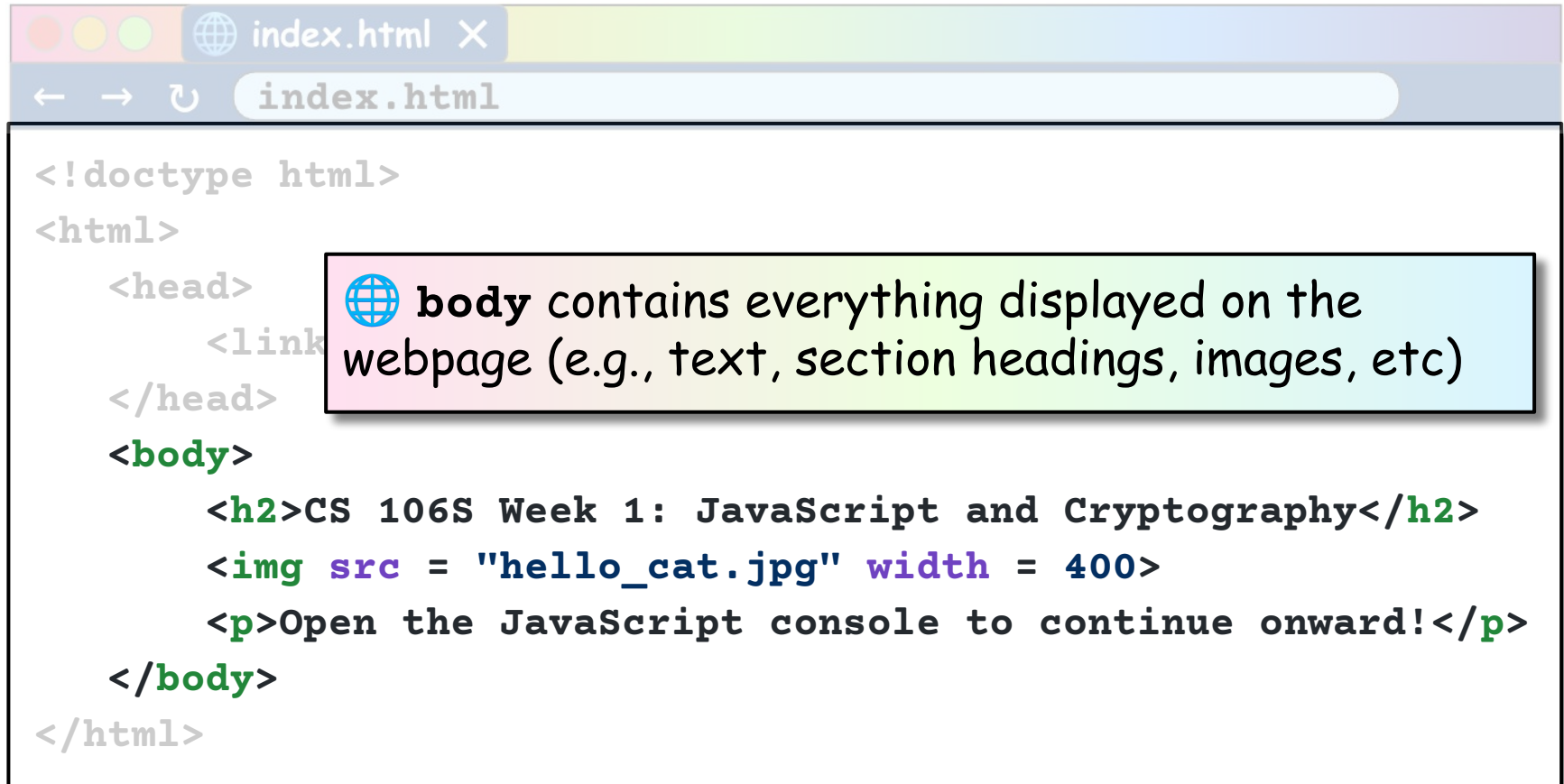
index.html

 **head** contains info not physically displayed on webpage (e.g., browser title, browser icon, any JavaScript or CSS style files to load)

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  </head>
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</html>
```

 **body** contains everything displayed on the webpage (e.g., text, section headings, images, etc)



Anatomy of an HTML File



```
<!doctype html>  
<html>
```

 Tags such as `<h2>` enclose each of the HTML elements. Typically have end tag (`</h2>`), but not always (`<img>`, `<br>` - line break)

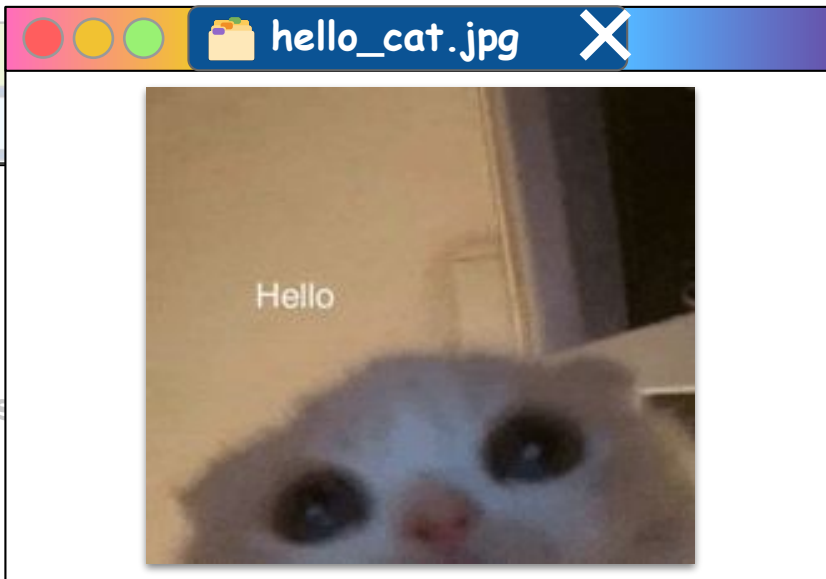
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</body>  
</html>
```



Anatomy of an HTML File



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Anatomy of an HTML File

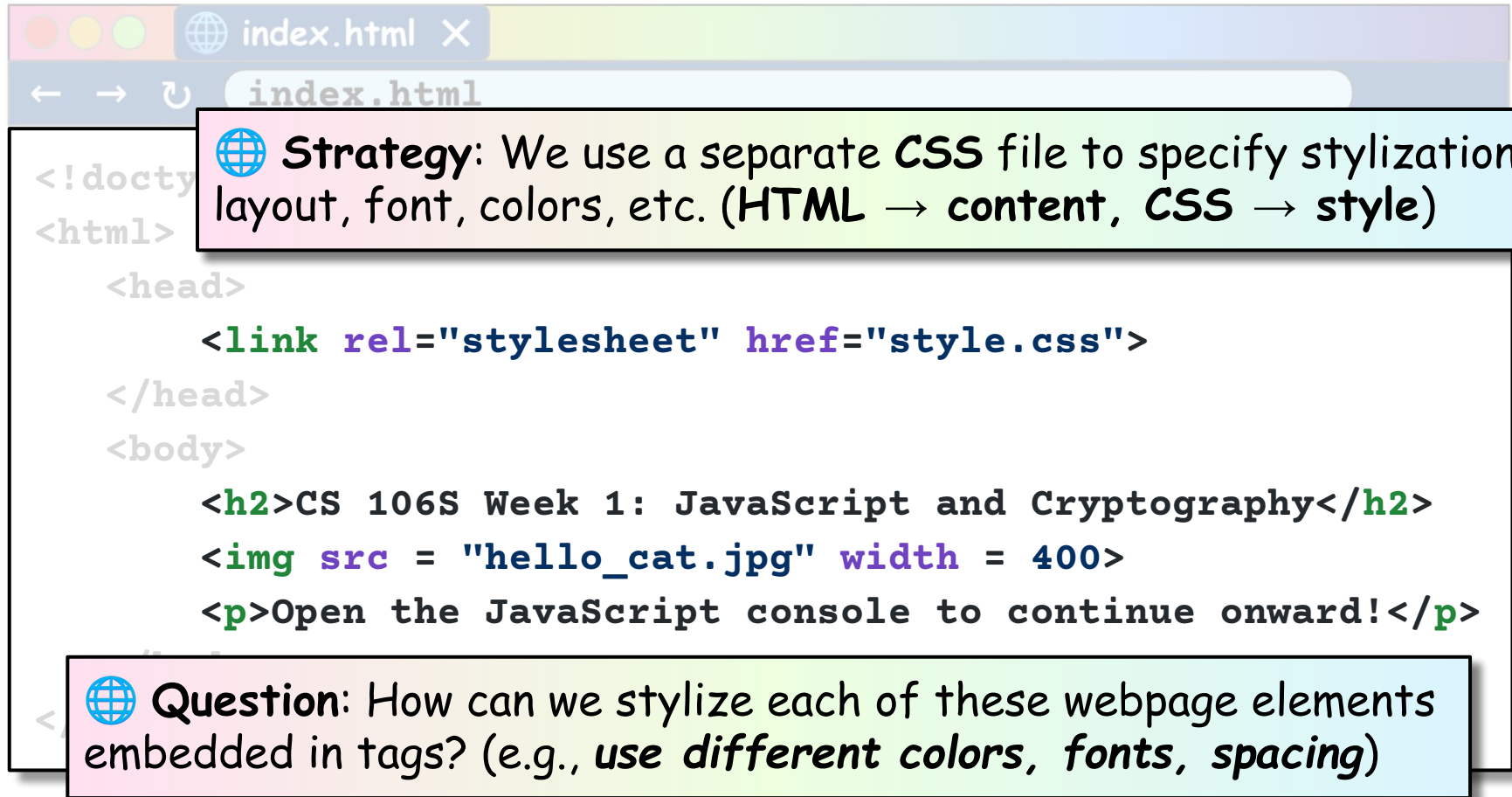
```
index.html X
index.html

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```

 **Question:** How can we stylize each of these webpage elements embedded in tags? (e.g., *use different colors, fonts, spacing*)



Anatomy of an HTML File

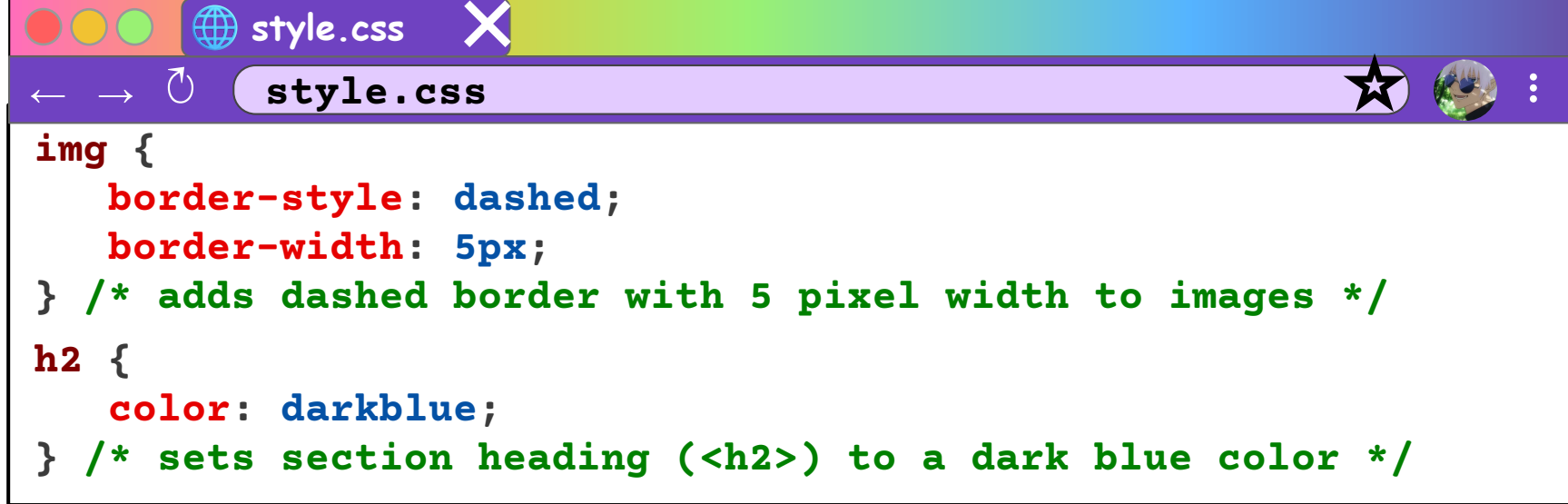


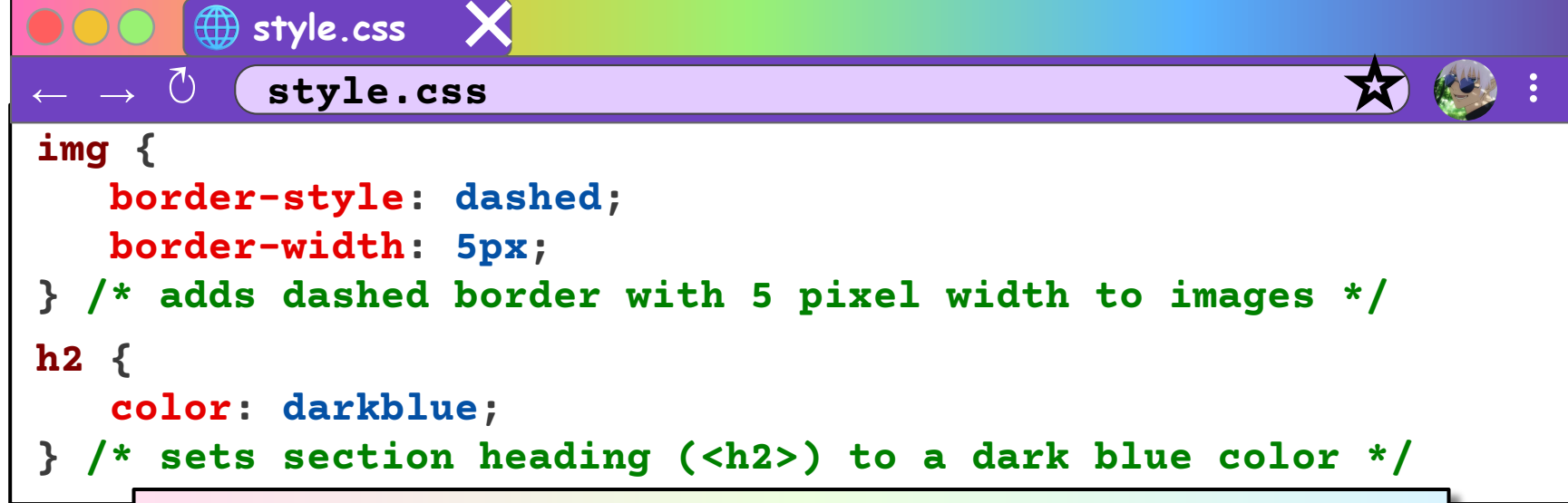
The screenshot shows a web browser window with the address bar displaying "index.html". A code editor overlay is positioned in front of the browser, showing the HTML code for "index.html". The code includes a DOCTYPE declaration, an HTML tag, a head section with a link to "style.css", and a body section containing an h2 heading, an img tag, and a paragraph. Two callout boxes provide additional context: one explains the strategy of using a separate CSS file, and the other poses a question about styling elements.

```
<!doctype>
<html>
  <head>
    <link rel="stylesheet" href="style.css">
  </head>
  <body>
    <h2>CS 106S Week 1: JavaScript and Cryptography</h2>
    <img src = "hello_cat.jpg" width = 400>
    <p>Open the JavaScript console to continue onward!</p>
  </body>
</html>
```

Strategy: We use a separate **CSS** file to specify stylization, layout, font, colors, etc. (**HTML** → content, **CSS** → style)

Question: How can we stylize each of these webpage elements embedded in tags? (e.g., *use different colors, fonts, spacing*)





The screenshot shows a web browser window with a single tab titled 'style.css'. The address bar also displays 'style.css'. The main content area shows the following CSS code:

```
img {  
    border-style: dashed;  
    border-width: 5px;  
} /* adds dashed border with 5 pixel width to images */  
  
h2 {  
    color: darkblue;  
} /* sets section heading (<h2>) to a dark blue color */
```

 In a CSS file, style rules are written in the form below:

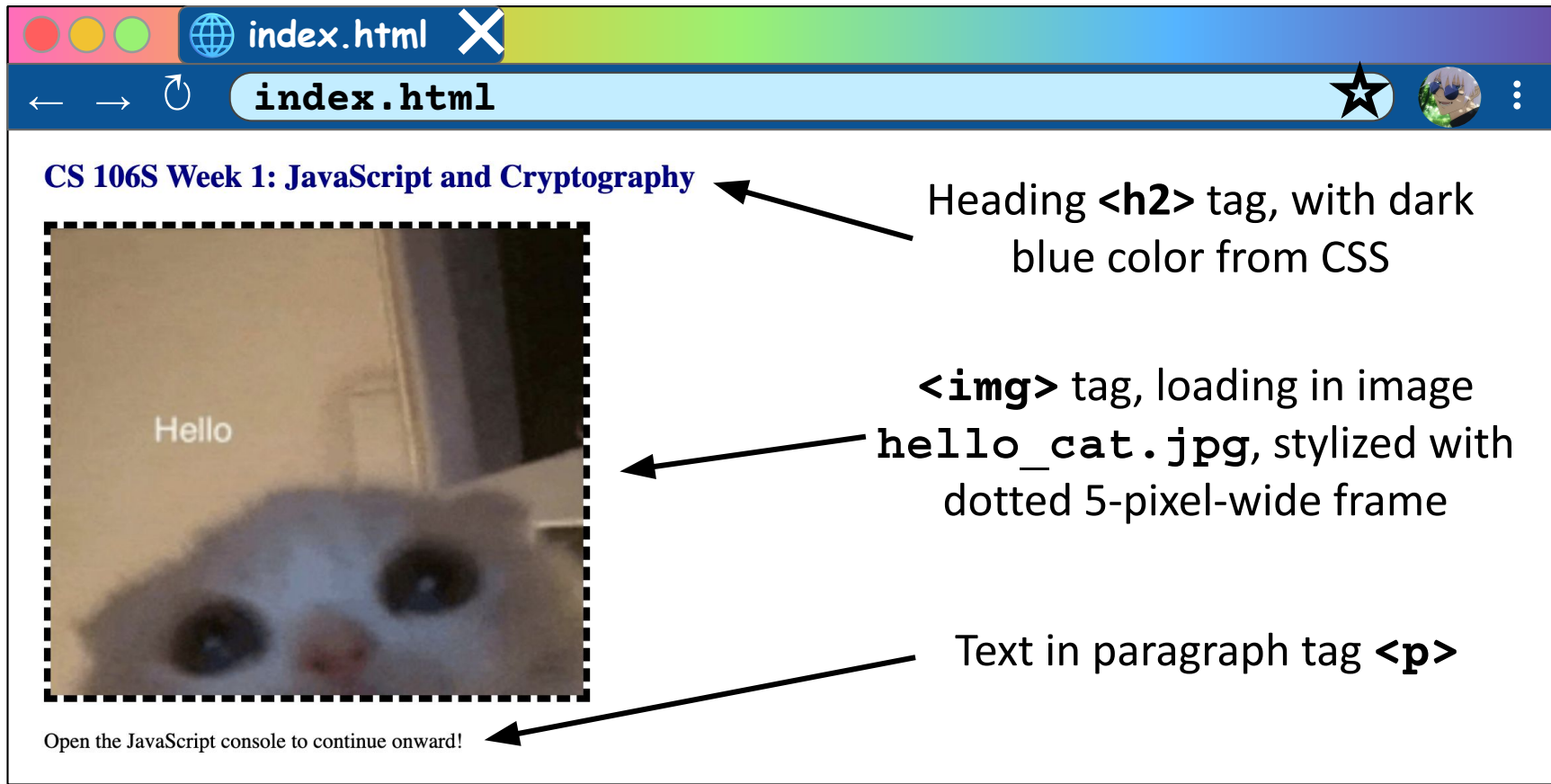
Which HTML element(s)
the styles should apply to,
e.g., all **** tags, the
id of a specific element

```
selector{  
    property-name: property-value  
    ... more pairs of CSS properties & values  
}
```

One of several CSS keywords specifying a style
detail, e.g., *font-family*, *font-size*, *color*,
background-color, *border-width*, etc.

Possible **values** depend on property name,
e.g., **color** (red, green), **font-family**
(serif, sans-serif, cursive, fantasy), etc.

HTML + CSS: Resulting Webpage, tada!



index.html

CS 106S Week 1: JavaScript and Cryptography

Hello

Open the JavaScript console to continue onward!

Heading `<h2>` tag, with dark blue color from CSS

`<img>` tag, loading in image `hello_cat.jpg`, stylized with dotted 5-pixel-wide frame

Text in paragraph tag `<p>`



What JavaScript does!

- ❑ JavaScript is a **powerful ‘interface’ with an HTML webpage**, enabling HTML elements to be programmatically accessed and modified.
 - ❑ Empowers **dynamic web apps** that respond to user interactions.
- ❑ To get a high-level sense of what JavaScript can do (before we dive into a starter tutorial on its features & syntax), it can:
 - ❑ Append, remove, or modify any HTML nodes i.e. tags/elements
 - Open pop-up window
 - Load new images
 - Toggle display to night mode
 - ❑ React to user events/actions with parts of the webpage, e.g., buttons
 - Button clicks
 - Hovering mouse over image
 - Keyboard key is pressed
- ❑ JS is now quite **general-purpose**! Has evolved beyond its web roots.



Onto the JS Tutorial! (on course website)



JavaScript Kick-off

Here's a simple Hello World program in JavaScript!

```
console.log("Hello World!");
```

JavaScript comments begin with a `//`, and while they're not executed as lines of actual code, they can help explain code and improve readability.

```
console.log("Hi there, my name is Ben!");  
// introducing myself to the computer, in a  
// friendly way in case they take over the world
```

JavaScript is a little quirky - in good and bad ways



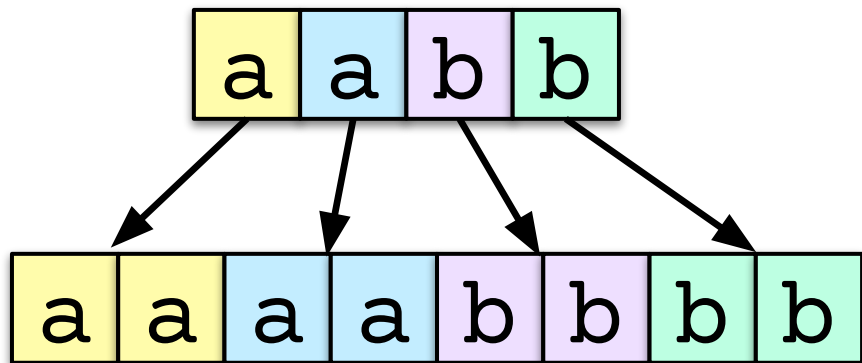
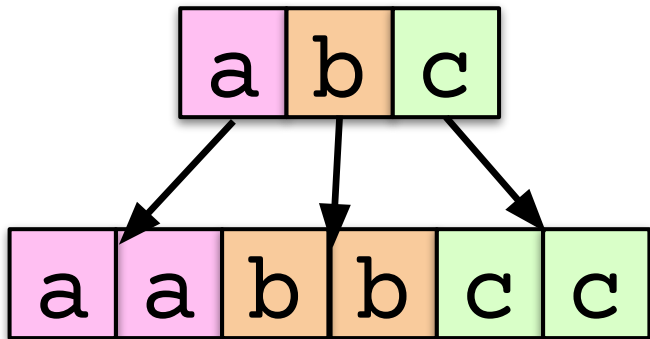


Overview: General JavaScript Syntax

Example: Consider a function `twins` which takes an input string and returns a copy of that string with each character repeated once. For example:



- ❑ `twins("abc")` *returns* "aabbcc"
- ❑ `twins("minnesota")` *returns* "mmiinnnneessoottaa"

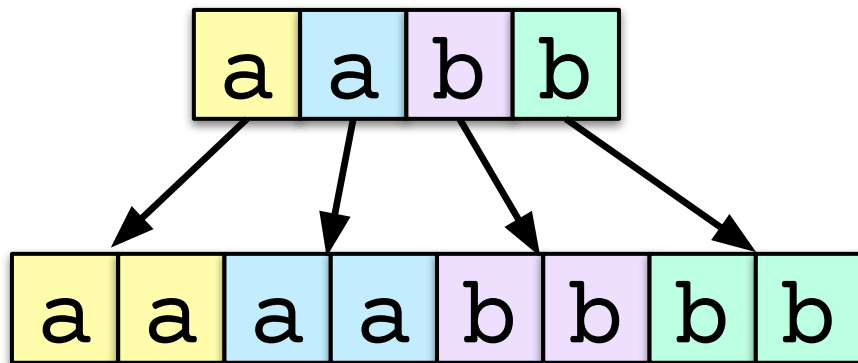
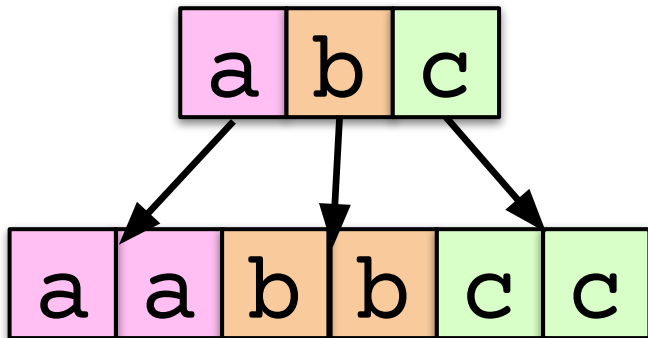




Overview: General JavaScript Syntax

```
function twins(str){ //JavaScript
  let result = "";
  for (let char of str){
    result += char + char;
  }
  return result;
}
```

```
def twins(str): #Python
  result = ""
  for char in str:
    result += char + char
  return result
```





Overview: General JavaScript Syntax

```
function twins(str){  
  let result = "";  
  for (let char of str){  
    result += char + char;  
  }  
  return result;  
}
```

- ❑ Python defines and segments blocks of code by indentation
- ❑ JavaScript uses **curly braces**, in the vein of C++, Java, and C.

```
function twins(str){  
  let result = "";  
  for (let char of str){  
    result += char + char;  
  }  
  return result;  
}
```

- ❑ In Python, having **parentheses** around statements / conditions in **for, while, if** loops is optional.
- ❑ In JavaScript, it's mandatory. Here's a **for/of** loop that iterates over the characters of a string.



Overview: General JavaScript Syntax

```
function twins(str){  
  let result = "";  
  for (let char of str){  
    result += char + char;  
  }  
  return result;  
}
```

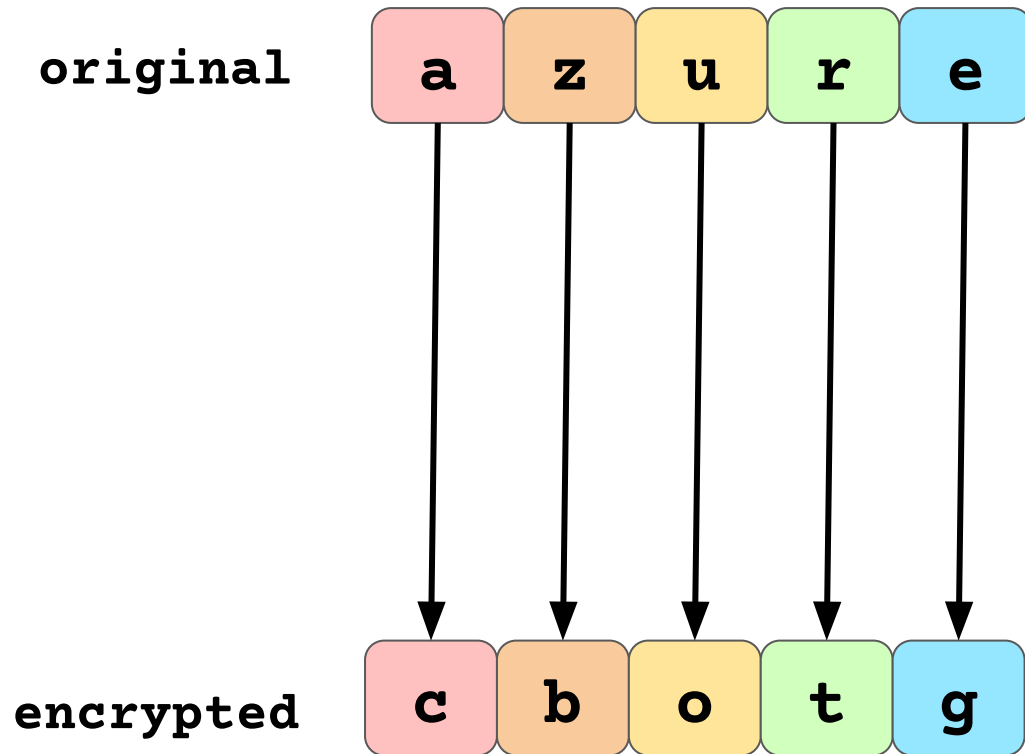
- ❑ In JavaScript, to declare variables of any type, we use the **let** keyword (or **const**).
- ❑ In Python, we could have simply written **result = ""**

```
function twins(str){  
  let result = "";  
  for (let char of str){  
    result += char + char;  
  }  
  return result;  
}
```

- ❑ Unlike C++, C, and Java, **semi-colons (;)** are optional at the end of statements.
- ❑ But you can include them if you'd like! :) It's a style or vibes-based decision ✨🔧



JavaScript Exercise: Caesar Cipher





JavaScript Exercise: Caesar Cipher

original

a z u r e

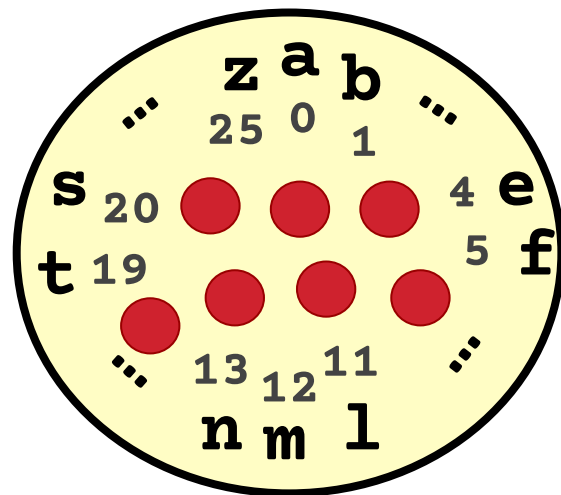
The core idea is that you shift each letter by some fixed number of alphabet positions, e.g., +2, wrapping around the end when needed.

encrypted

c b o t g

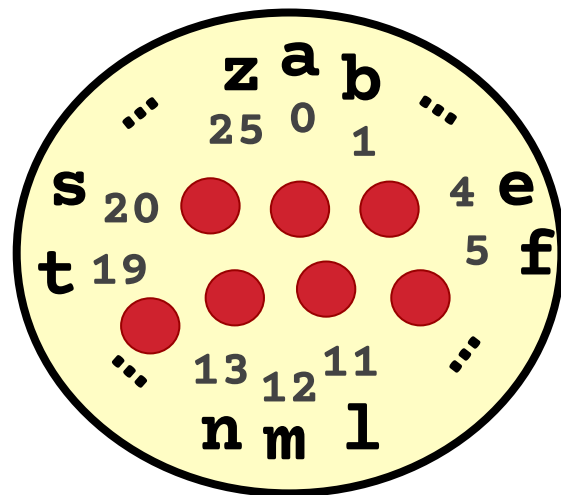
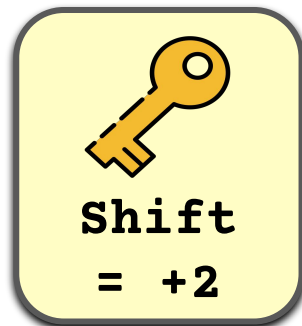
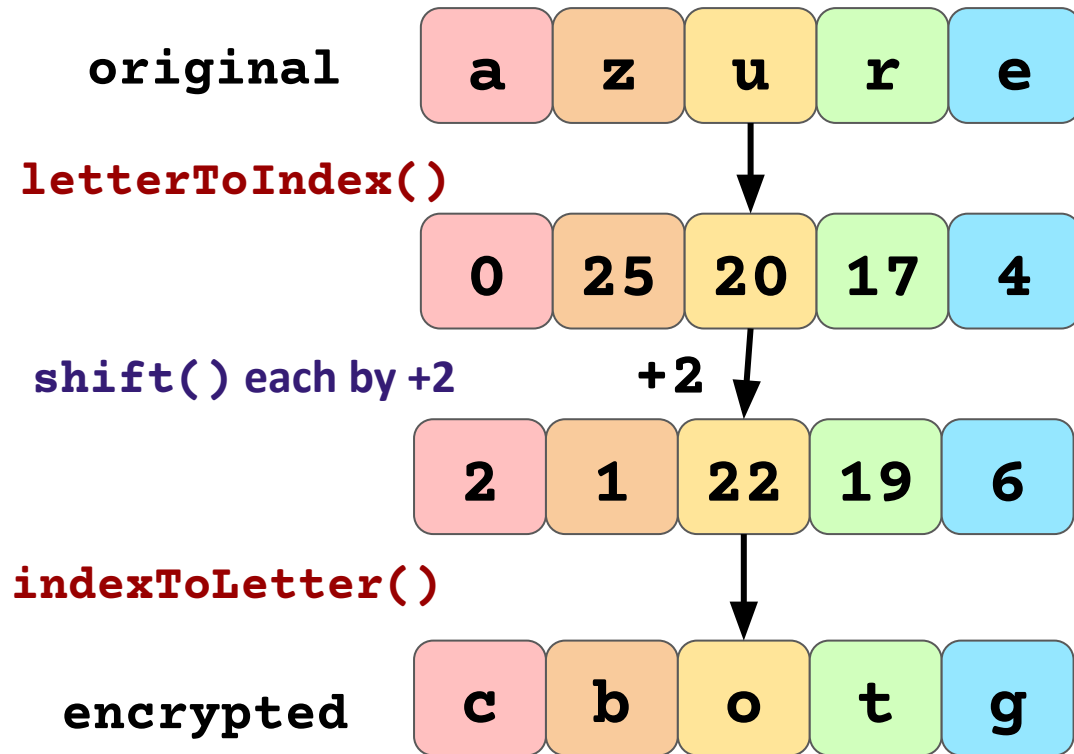


Shift
= +2



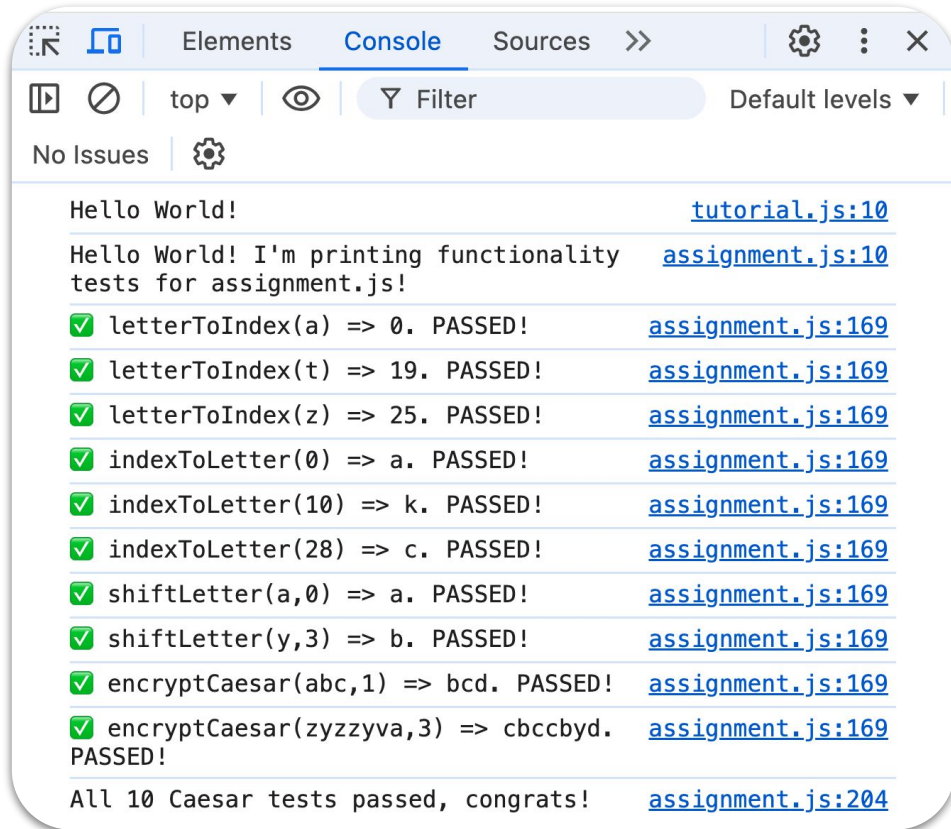
A Little Caesars Pizza!

🔑 JavaScript Exercise: Caesar Cipher



A Little Caesars Pizza!

Note: After editing the `.js` file, make sure to click **File** -> **Save** in VSCode, and refresh the page, for the edits to reflect in the console!



The screenshot shows a web browser's developer console with the 'Console' tab selected. The console displays the following output:

```
Hello World! tutorial.js:10  
Hello World! I'm printing functionality tests for assignment.js! assignment.js:10  
✓ letterToIndex(a) => 0. PASSED! assignment.js:169  
✓ letterToIndex(t) => 19. PASSED! assignment.js:169  
✓ letterToIndex(z) => 25. PASSED! assignment.js:169  
✓ indexToLetter(0) => a. PASSED! assignment.js:169  
✓ indexToLetter(10) => k. PASSED! assignment.js:169  
✓ indexToLetter(28) => c. PASSED! assignment.js:169  
✓ shiftLetter(a,0) => a. PASSED! assignment.js:169  
✓ shiftLetter(y,3) => b. PASSED! assignment.js:169  
✓ encryptCaesar(abc,1) => bcd. PASSED! assignment.js:169  
✓ encryptCaesar(zyzzyva,3) => cbcbyd. PASSED! assignment.js:169  
All 10 Caesar tests passed, congrats! assignment.js:204
```

All tests should pass after `encryptCaesar()` is successfully implemented!

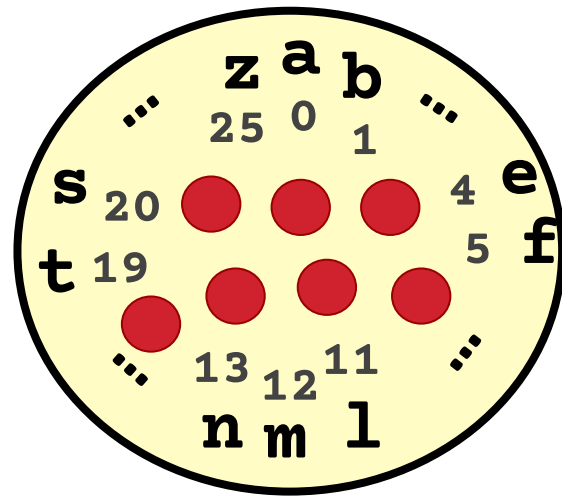
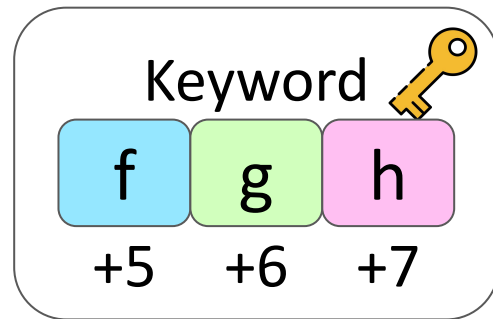
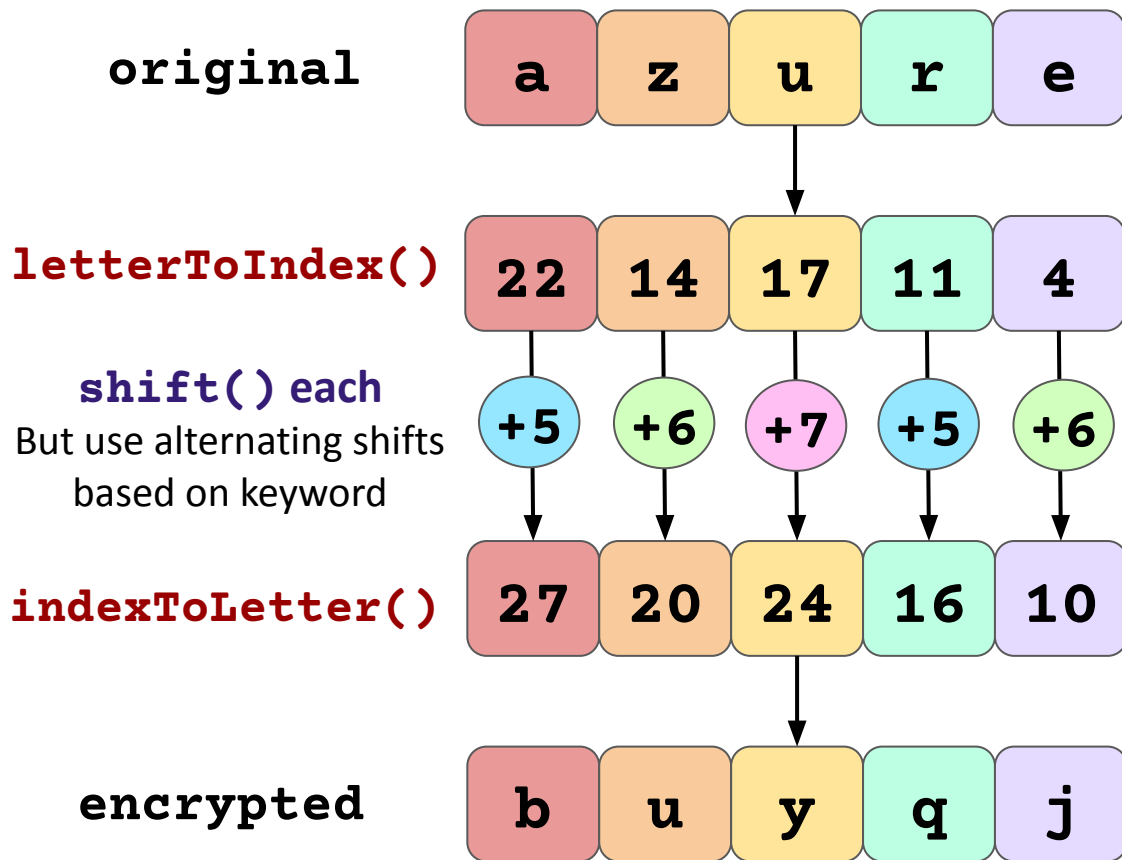
Solution code available on website right after class :)

To test Vigenère (optional!), go to the bottom of `assignment.js`, and uncomment the line:

```
// testVigenere()
```

Bonus: Vigenère Cipher

Remark: It's like having multiple Caesar ciphers in one encryption!
A rotating set of keys.





And lastly, the check-off form!

To receive **attendance credit each class**, you'll fill out a brief check-off form! (~ 5 min or less to complete).



Week 1 Check-Off (CS106S, Spring 25 🌱)

👋 Hello! Please note that completing this form is how we check attendance :)
Really looking forward to a fun quarter ahead!

bbyan@stanford.edu [Switch account](#) 

* Indicates required question

Email *

Your email

Full Name (First + Last) *

Your answer



And lastly, the check-off form!

To receive **attendance credit each class**, you'll fill out a brief check-off form! (~ 5 min or less to complete).

Click me!

Check-Off Form

For today, click the "**Check Off Form**" button on the website for Week 1. Or access the case-sensitive link below!

<https://tinyurl.com/cs106s-aut26-w1-checkoff>



Week 1 Check-Off (CS106S, Spring 25 🌱)

👋 Hello! Please note that completing this form is how we check attendance :)
Really looking forward to a fun quarter ahead!

bbyan@stanford.edu [Switch account](#) 

* Indicates required question

Email *

Your email

Full Name (First + Last) *

Your answer



```

Welcome to

EEEEEE LL      IIII ZZZZZZ AAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LL      II      ZZ  AAAAAA
EE      LL      II      ZZ  AA  AA
EEEEEE LLLLLL IIII ZZZZZZ AA  AA

Eliza is a mock Rogerian psychotherapist.
The original program was described by Joseph Weizenbaum in 1966.
This implementation by Norbert Landsteiner 2005.

ELIZA: Is something troubling you ?
YOU:  Men are all alike.
ELIZA: What is the connection, do you suppose ?
YOU:  They're always bugging us about something or other.
ELIZA: Can you think of a specific example ?
YOU:  Well, my boyfriend made me come here.
ELIZA: Is it important to you that your boyfriend made you come here ?
YOU:  He says I'm depressed much of the time.
ELIZA: I am sorry to hear that you are depressed.
YOU:  It's true. I am unhappy.
ELIZA: Can you explain what made you unhappy ?
YOU:
  
```

Have an amazing week, and welcome to CS106S!

Please feel free to reach out with any questions! (bbyan@stanford.edu)

